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Endogenous corazonin signaling modulates the post-mating switch in behavior and physiology in females of the brown planthopper and *Drosophila*

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eLife Assessment

This **important** study presents **convincing** evidence that uncovers a novel signaling axis impacting the post-mating response in females of the brown planthopper. The findings open several avenues for testing the molecular and neurobiological mechanisms of mating behavior in insects, and in the revised version the authors provide further evidence supporting their conclusions.

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Abstract

Mating in insects typically triggers a post-mating response (PMR) in females, characterized by reduced receptivity to re-mating and increased oviposition, which ensures numerous and viable offspring and male paternity. This PMR is induced by male seminal factors, such as sex peptide in *Drosophila melanogaster*, as well as intrinsic female signaling components. The latter signaling remains poorly understood in most insects, including the devastating rice pest, the brown planthopper (BPH) *Nilaparvata lugens*. Here, we show that the neuropeptide corazonin (CRZ) and its receptor (CrzR) are critical for the PMR in female BPHs. Peptide injection, RNAi knockdown, and CRISPR/Cas9 mutagenesis confirm that intact CRZ signaling reduces re-mating frequency and increases ovulation in mated BPH females. The CrzR is highly expressed in the female reproductive tract, and CrzR knockdown phenocopies CRZ diminishment. Importantly, female CRZ/CrzR signaling is required for male seminal factors, such as the peptide maccessin, to induce the PMR; with disrupted *CrzR* signaling, injection of seminal fluid or maccessin fails to reduce female receptivity. Notably, CRZ is not produced in male accessory glands (MAGs) of BPHs and thus not transferred during copulation. We furthermore demonstrate that also in *D. melanogaster* disrupted CRZ signaling increases female re-mating and reduces oviposition, while CRZ injection suppresses virgin receptivity and increases oviposition. Finally, we detected no CRZ in the MAG of *D. melanogaster*, supporting its role as an endogenous signal in the female PMR also in this species. In summary, our findings reveal a conserved role of endogenous CRZ signaling in regulating the female PMR and demonstrate that female CRZ signaling and male-derived signals cooperate to induce post-mating transitions in BPHs and *D. melanogaster*. CRZ is a paralog of the peptide gonadotropin-releasing hormone, known to regulate reproduction in vertebrates, including humans, suggesting evolutionary conservation of an ancient function.

Introduction

Reproductive behavior in animals is influenced by a complex interplay between external sensory inputs and internal states, and is regulated by neuronal and hormonal signaling. In insects, males typically initiate mating through courtship, while females determine acceptance by integrating external sensory cues with their internal state. Consequently, female reproductive activity is modulated by multiple signaling pathways [1–8]. Importantly, in insects, successful copulation commonly induces profound physiological and behavioral changes in the females [5,9–18].

Following their first mating, insect females typically reject further courtship attempts and initiate oviposition. This distinct shift in behavior and physiology is referred to as a post-mating response (PMR) [5–7,19–23]. Induction of a PMR has been documented across diverse insect taxa, for example in *D. melanogaster*, mosquitos *Anopheles gambiae*, *Aedes aegypti*, crickets (*Gryllodes sigillatus*), and the brown planthopper (BPH), *Nilaparvata lugens* [6,10,15,18,21,24–28].

Induction of a PMR commonly requires the transfer of specific seminal fluid components during copulation. In *D. melanogaster*, seminal fluid peptides, including sex peptide (SP) and DUP99B produced in male accessory glands (MAG), enter the female reproductive tract, bind to specific receptors on sensory neurons, and activate signaling pathways that drive post-mating behavior and physiology [6,13,14,18,29]. Additional factors such as Acp26Ab and ovulin (Acp26Aa) are also transferred with the seminal fluid and modulate female reproductive physiology in *D. melanogaster* [30–33]. Notably, it has been shown that the PMR-inducing substances, produced in the MAG of various insects, are highly species-specific and not even conserved across related taxonomic groups [24,34].

While MAG-derived sex peptide (SP) and its receptor (SPR) are key regulators of the female post-mating response (PMR) in *D. melanogaster* [5–7,16], additional signaling pathways intrinsic to females also modulate this process. Interneurons within abdominal neuromeres of the ventral nerve cord (VNC) promote receptivity to males by signaling via myoinhibitory peptide (MIP) to brain circuits [35]. Furthermore, the neuropeptide diuretic hormone 44 (DH44) modulates female sexual receptivity through a sex-specific brain circuit; mating suppresses DH44 signaling, thereby reducing receptivity [1]. During sexual maturation, leucokinin (LK)-expressing neurons (ABLKs) in the abdominal ganglion, downstream of SP-activated circuitry, suppress female receptivity [36]. In addition to changes in receptivity, the *D. melanogaster* PMR includes a shift in diet preference toward yeast consumption. This altered feeding behavior is mediated by specific peptidergic neurons (ALKs) in the brain through LK signaling [12].

Although the mechanisms and neuronal pathways underlying the post-mating response (PMR) in *D. melanogaster* are well characterized [3,4,6,7,37–40], this process remains poorly understood in most non-model insects. A prominent example is the brown planthopper (BPH), *N. lugens* (Hemiptera). As a monophagous rice pest, the BPH exhibits robust reproductive capacity and remarkable environmental adaptability[41]. It has also developed high resistance to diverse chemical pesticides [42–44]. Consequently, elucidating reproductive physiology and behavior in BPHs is crucial for developing novel control strategies.

In BPHs, the recently identified MAG-derived peptide maccessin (macc) is transferred during copulation and triggers a PMR in females, although its receptor remains unknown [24]. Notably, the same study demonstrated that a second peptide, ion transport-like peptide 1 (ITPL-1), also modulates the PMR. Whereas macc is male-specific, ITPL-1 is produced both in the MAG and endogenously in females, where it reduces female receptivity to courting males [24]. Given its endogenous female expression, ITPL-1 likely functions as an intrinsic signaling component in the PMR of BPHs, analogous to the role of myoinhibitory peptide (MIP) and DH44 in *D. melanogaster* females [1,35].

Here, we explored the role of another neuropeptide, corazonin (CRZ), in the reproductive behavior of BPHs. In *D. melanogaster* males, CRZ is released from interneurons of the ventral nerve cord (VNC) and acts on serotonergic projection neurons to regulate ejaculation and copulation duration [45], mitigates mating-induced heart rate acceleration for physiological recovery [46] and influences post-mating dietary preference [12]. Surprisingly, the role of CRZ signaling in female reproduction remains unexplored. To address this gap, we investigated the role of CRZ in female reproduction, including the PMR, in BPHs and *D. melanogaster*.

Here, we demonstrate that CRZ induces a post-mating switch in receptivity in females of both species. CRZ injection in virgin females reduces receptivity to courting males and increases oviposition, while *Crz* knockdown or knockout in mated females restores receptivity to re-mating and decreases oviposition. Notably, CRZ is not produced in the male accessory gland (MAG) of either species and thus the peptide is not transferred during copulation, but acting as an endogenous female peptide signal. In BPHs, the CRZ receptor (CrzR) is highly expressed in the female reproductive tract and both CRISPR-Cas9-mediated *CrzR* knockout and RNAi-mediated knockdown disrupt the PMR, phenocopying CRZ deficiency. Critically, receptivity changes in virgin BPH female, whether induced by MAG extract (containing seminal fluid proteins) or CRZ injection, depend on presence of the CrzR. Collectively, our data demonstrate that endogenous female CRZ signaling and male seminal cues are both required to coordinate post-mating changes in physiology and behavior in females of both insects. CRZ is a paralog of the peptide gonadotropin-releasing hormone, known to regulate reproduction in vertebrates, including humans, suggesting evolutionary conservation of an ancient function.

Results

CRZ signaling reduces receptivity to courting males and stimulates oviposition in *N. lugens* females

While numerous studies have established roles of the neuropeptide CRZ in regulating aspects of male reproduction across diverse insect species [47–53], research addressing its potential functions in female reproduction remains conspicuously lacking.

This study investigates the role of CRZ in female reproduction, starting with the brown planthopper (BPH), *Nilaparvata lugens*, a major agricultural pest. Initially, the *Crz* gene (GenBank accession: AB817247.1) was cloned based on transcriptome data [54]. The deduced mature CRZ sequence in BPH, pQTFQYSRGWTNamide, is identical to that reported for most studied insect species (Figure 1-figure supplement 1A). Consistent with other known CRZ precursors, the BPH precursor encodes a single CRZ peptide starting at the first amino acid of the propeptide (Figure 1-figure supplement 1A and B). A comparison of the organization of selected CRZ precursor genes is shown in Figure 1-figure supplement 1B. Genomic analysis revealed that the CRZ precursor is encoded by two exons separated by a single intron (Figure 1-figure supplement 1B). The mature CRZ peptide and a scrambled control peptide (sCRZ) were synthesized for pharmacological experiments.

First, we investigated the role of CRZ signaling in the reproductive behavior and PMR of female BPHs. The PMR in BPHs, is characterized by reduced receptivity to courting males, including display of rejection behaviors, and increased oviposition [24]. We found that injection of CRZ significantly reduced receptivity of virgin female BPHs to courting males (Figure 1A and B). Furthermore, virgin females injected with CRZ actively rejected courting males, exhibiting a distinct rejection behavior characterized by ovipositor extrusion and kicking or fleeing. This behavior was not observed in control females injected with sCRZ (Figure 1 Supplementary Video 1). Notably, the CRZ-induced mating refusal behavior in virgin females was no longer observed 24 hours after injection (Figure 1-figure supplement 2A). While injection of CRZ peptide did not stimulate egg production in virgin BPHs (Figure 1-figure supplement 2B), it significantly promoted oviposition in mated females (Figure 1C).

To further investigate CRZ action, we performed RNA interference (RNAi) by injecting double-stranded *Crz* RNA (*dsNlCrz*) into female BPHs, which efficiently reduced *Crz* expression (Figure 1-figure supplement 2). In *N. lugens*, female receptivity is transiently suppressed after mating and does not begin to recover until at least 72 h after the first mating[24]; therefore, remating assays were performed at 48 h post-mating as shown in Figure 1D. Knockdown of *Crz* resulted in

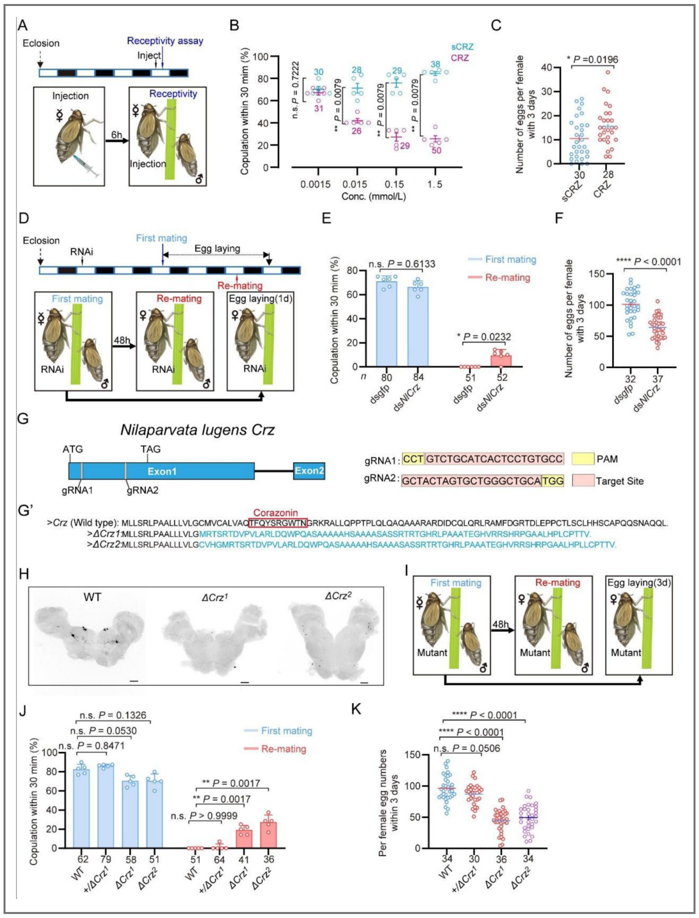


Figure 1. CRZ signaling affects the post-mating response (PMR) and oviposition in female *N. lugens*.

Wild-type males were used in all behavioral assays. (A) Experimental design for panels B-C. The white/black segments on the experimental time lines denote light/dark periods, respectively. (B) Receptivity of virgin females 6 h post-injection, using different doses of CRZ (sCRZ as control), each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng/10 ng/1 ng/0.1 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. The graph shows percentage females copulating within 30 min. **$P < 0.01$ vs. control; ns (non-significant): $P > 0.05$ (Mann-Whitney test). The small circles denote the number of replicates, ≥5 insects/replicate; the numbers above the curves denote total number of animals. (C) Eggs laid per mated female 12 h post-copulation. We used the highest CRZ dose from panel B (sCRZ, control). Each mated female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. *$P < 0.05$ (Student's t-test). Data: mean ± s.e.m (≥4 biological replicates, ≥6 insects/replicate). The numbers below the bars denote total number of animals. (D) Experimental design for panels E-F. (E) Receptivity of virgin (1st mating) and mated females after *Crz*-RNAi (*dsgfp* as control). Graph displays the percentage of females copulating within 30 min. The small circles denote the number of replicates, ≥7 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (F) Total numbers of eggs laid per female over 72 h after *Crz*-RNAi (*dsgfp* as control). ****$P < 0.0001$ (Student's *t*-test). The small circles and the numbers below the bars denote total number of animals, (≥4 biological replicates, ≥8 insects/replicate). Data are shown as mean ± s.e.m. (≥3 biological replicates, ≥8 insects/replicate). (G) Schematic diagram of the *Crz* gene and sgRNA (single-guide RNA) design for *Crz* mutant production. The *Crz* gene consists of 2 exons, and ATG and TAG are located on Exon 1. The target site of the 20 bp sgRNA on Exon 1 is highlighted in pink. The PAM sites are indicated in yellow. The two knockout strains ΔCrz^1 and ΔCrz^2 were unable to produce mature CRZ peptides. (G') The amino acid sequences obtained by translating the mutated base sequences. (H) Absence of CRZ immunoreactivity in homozygous ΔCrz^1 and ΔCrz^2 mutants compared to wild type (WT) expression. Scale bars: 50 µm. (I) Experimental design for panels J-K. (J) Receptivity of virgin/mated females across genotypes. No effect was seen on virgin receptivity, only the re-mating was affected. The small circles denote the number of replicates; ≥6 insects/replicate, the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (K) Eggs laid 72 h post-mating. The numbers below the bars denote total number of animals., ≥4 biological replicates, ≥8 insects/replicate. Data are shown as mean ± s.e.m. ****$P < 0.0001$; Student's *t* test.

increased receptivity with 10% of the mated females mating again, and significantly reduced oviposition to approximately 60% of control levels (Figure 1E and F). In contrast, *Crz* knockdown had no effect on the receptivity of virgin females (Figure 1E).

Since RNAi-mediated knockdown of *Crz* in female BPHs left a residual level of gene expression (Figure 1-figure supplement 2), we generated two *Crz* deletion alleles, ΔCrz^1 and ΔCrz^2, using CRISPR/Cas9 (Figure 1G-G′) to achieve a complete ablation. Both alleles harbor deletions eliminating the mature CRZ peptide coding sequence (Figure 1G′′). Sanger sequencing confirmed that the deletions compromised essential triplet codons within the *Crz* open reading frame (Figure 1-figure supplement 4), thereby efficiently disrupting CRZ peptide expression (Figure 1G′′ and H). Immunohistochemical analysis further validated the mutants, revealing a complete absence of anti-CRZ immunolabeling in homozygous ΔCrz^1 and ΔCrz^2 females (Figure 1H). Homozygous ΔCrz^1 and ΔCrz^2 virgin females are fully viable and fertile. They exhibit a courtship vigor similar to controls and their initial mating success rate with wild-type males exceeds 60% (Figure 1I and J). However, 25% of the mated mutant females were receptive to further mating (Figure 1J), and oviposition was significantly reduced to approximately 50% of the wild-type levels (Figure 1K). To further test the *Crz* mutants, we also analyzed mating and oviposition behaviors in *Crz* heterozygous ($+/\Delta Crz^1$) females. Notably, neither the remating rate nor egg production in heterozygous females differed significantly from those of wild-type controls (Figure 1J and K). These results indicate that the *Crz* loss-of-function phenotype is recessive, and that a single functional copy of *Crz* is sufficient to sustain a normal post-mating response (PMR). Collectively, these results demonstrate that diminished CRZ signaling severely perturbs the PMR in female BPHs, seen as increased post-mating receptivity and reduced egg-laying. Importantly, the *Crz* deletion abolished the onset of typical post-mating rejection behavior. We, thus, conclude that endogenous CRZ signaling is important for the induction of a PMR in female BPHs.

CRZ is produced by similar neurons in both sexes of BPHs and is absent from the male accessory gland

To further understand how CRZ regulates the PMR in BPHs, we examined the expression of CRZ and *Crz* across different tissues in males and females. The *Crz* mRNA expression was examined in various BPH tissues using both RT-PCR and qPCR (Figure 2A). We detected *Crz* mRNA expression primarily in the central nervous system (CNS), regardless of the primer set or detection method used, while expression was nearly undetectable in other tissues, including reproductive organs of both sexes (Figure 2B and C).

A previous study reported CRZ in four male-specific interneurons within the abdominal ganglia of *D. melanogaster* [47]. To determine whether CRZ exhibits sexually dimorphic expression in BPHs, we performed immunolabeling using an anti-CRZ antiserum. This labeled four pairs of neurons in the protocerebrum of the brain (Figure 2D and E) and one pair of neurons in the subesophageal ganglion in both males and females (Figure 2D and E). Since CRZ immunolabeled neurons in brains of most, if not all, insects studied include three or more pairs of lateral neurosecretory cells [55,56], we suggest that CRZ signaling regulating the PMR may be hormonal (at least in part). Importantly, no CRZ-immunopositive neurons were detected in the abdominal ganglia of either sex (Figure 2D and E), and thus there is no evidence for sex-specific expression of CRZ in neurons in BPHs. This is in contrast to *D. melanogaster*, where we confirmed, as previously reported [45], the presence of CRZ-immunolabeled neurons specifically in the abdominal ganglia of males (Figure 2 - figure supplement 1A).

To exclude that CRZ acts as a seminal fluid peptide transferred to females during copulation, similar to sex peptide (SP) in *D. melanogaster* and macc in BPHs [24], we specifically examined CRZ expression in reproductive tissues of both sexes. However, we observed no CRZ immunolabeling in the male reproductive organs (Figure 2F1 - F3) or female reproductive organs (Figure 2G1 – G3). This aligns with our transcriptional *Crz* data shown above (Figure 2B and C). Furthermore, *Crz* mRNA and CRZ protein were absent from transcriptomic and proteomic datasets

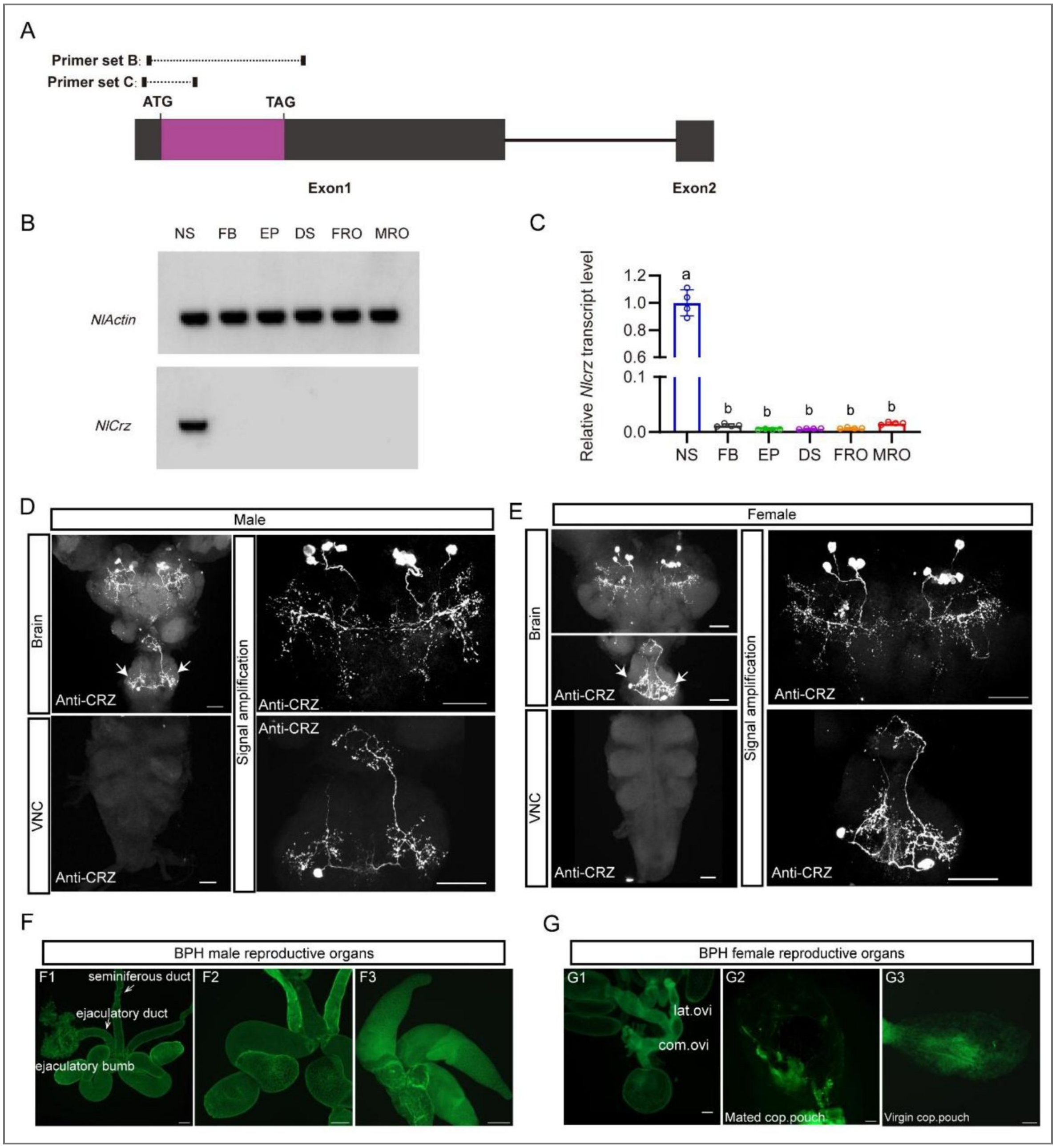


Figure 2. Expression of *Crz* and CRZ in the nervous system of the brown planthopper (BPH).

(A) Schematic of primer design for *Crz* detection (primer sequences provided in Table 1). (B) RT-PCR analysis of *Crz* mRNA levels in various BPH tissues. Actin served as the loading control. Tissues: Nervous system (NS), fat body (FB), Epidermis (EP), digestive system (DS), female reproductive organs (FRO), male reproductive organs (MRO). Note: Tissues except reproductive organs were pooled from both sexes. (C) Relative *Crz* expression in female *N. lugens* tissues determined by RT-qPCR (abbreviations as in B). Expression was normalized to *Actin* and *18SrRNA* and expressed relative to the mean value of the highest-expressing tissue (set to 1). Data are shown as mean ± s.e.m. Groups that share at least one letter are statistically indistinguishable. One-way ANOVA followed by Tukey's multiple comparisons test. (D and E) Immunolocalization of CRZ peptide in the nervous system of male (D) and female (E) BPHs. Four pairs of CRZ neurons in the brain and one pair in the subesophageal ganglion are shown. Note that some of the CRZ neurons are likely to be neurosecretory cells with axon terminations in the corpora cardiaca, others may be interneurons. The extensive branches within the brain suggest CRZ signaling in brain circuits. VNC, ventral nerve cord. Scale bars: 50 μm. (E) CRZ immunolabeling in the nervous system of female BPHs. Scale bar: 50 μm. (F and G) Lack of CRZ immunoreactivity in BPH reproductive organs. (F1-F3) Male reproductive organs (MRO): testes (test). (G1-G3) Female reproductive organs (FRO): lateral oviduct (Lat. ov), common oviduct (Com. ov). Scale bars: 100 μm.

of the male accessory glands (MAGs) in BPHs [24]. Collectively the available data suggest, that CRZ is not produced by the MAG and is consequently unlikely to be transferred to females as a seminal fluid component.

The CRZ receptor is essential for mediating the post mating response in female BPHs

Given our findings implicating CRZ in modulating the PMR in female BPHs, we identified the CRZ receptor (*CrzR*) gene in this species; it encodes a structurally conserved GPCR orthologous to other insect *CrzR*s (Figure 3-supplemental figure 1). We next explored the effect of knocking down the *CrzR* on the PMR in BPHs. RNAi-mediated knockdown of the receptor (*dsCrzR* injection) resulted in 15% of the mated females re-mating, a phenotype not seen in *dsgfp*-injected controls (Figure 3A and B) and in significantly reduced oviposition (Figure 3C). Importantly, CRZ injection failed to diminish receptivity in *CrzR*-knockdown virgins (Figure 3D and E). Furthermore, MAG extract (with seminal fluid proteins) injection into *dsCrzR*-treated virgins did not affect receptivity (Figure 3F).

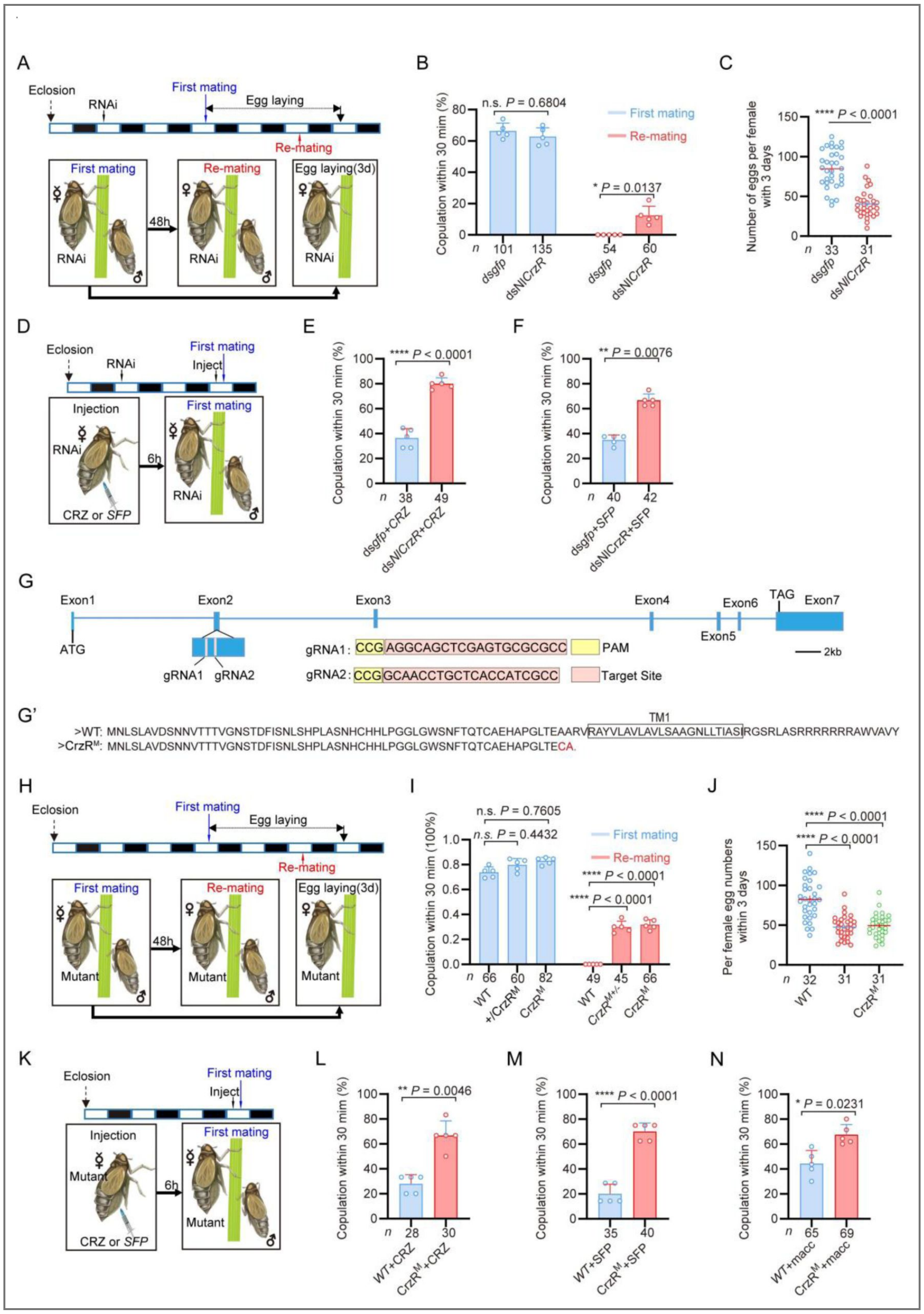


Figure 3. **The CRZ receptor (*CrzR*) is essential for the female PMR and is required for action of MAG-derived factors.**

(A) Experimental design for B and C. (B) Receptivity after *CrzR*-RNAi (injection of ds*CrzR*; ds*GFP* as control). Percentage females copulating within 30 min. The small circles denote the number of replicates, ≥9 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (C) Numbers of eggs laid within 72 h post-mating (*dsCrzR* females × WT males). Data are shown as mean ± s.e.m. ****$P < 0.0001$ (Student's *t*-test). The numbers below the bars denote total number of animals, ≥4 biological replicates, ≥7 insects/replicate. (D) Experimental design for E-F. (E) Receptivity after CRZ injection in *CrzR*-knockdown virgins (*dsGFP* + CRZ control). Graph shows percentage females copulating within 30 min. Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 10 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. The small circles denote the number of replicates, ≥7 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (F) Receptivity after MAG extract (with seminal fluid proteins) injection in *CrzR*-knockdown virgins (*dsGFP* + SFP control). Percentage copulating within 30 min. The small circles denote the number of replicates, ≥7 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (G) *CrzR* targeting strategy for generation of mutants. *Top:* Genomic structure (ATG/TAG in Exons 2/7). sgRNA target (pink), PAM (yellow). *Bottom:* CRISPR alleles. (G') Translated receptor mutant sequences. (H) Experimental design for I-J. (I) Receptivity of *CrzR* mutants compared to wild type animals (WT). Graph displays the percentage of females copulating within 30 min. The small circles denote the number of replicates, ≥9 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (J) Number of eggs laid by mutants mated with WT males within 72 h. The numbers below the bars denote total number of animals, ≥4 biological replicates, ≥7 insects/replicate. Data are shown as mean ± s.e.m. ****$P < 0.0001$; Student's *t* test. (K) Experimental design for L-M. (L) Receptivity after CRZ injection in *CrzR* mutants compared to WT. Graph displays the percentage of females copulating within 30 min. Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 10 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. The small circles denote the number of replicates, ≥5 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (M) Receptivity after SFP injection in *CrzR* mutants compared to WT. Graph displays the percentage of females copulating within 30 min. The small circles denote the number of replicates, ≥7 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (N) Receptivity after macc injection in *CrzR* mutants compared to WT. Graph displays the percentage of females copulating within 30 min. The small circles denote the number of replicates, ≥10 insects/replicate; the numbers below the bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant.

To further confirm the role of the *CrzR* in the PMR of BPHs, we applied CRISPR/Cas9 genome editing to mutate the *CrzR* gene and constructed a homozygous mutant strain of *CrzR* ($CrzR^M$) by genetic crosses (Figure 3G-3G'', Fig. 3 - figure supplement 2A). We found that the developmental time and life span of the female *CrzR* mutants, like the *Crz* mutants, were slightly extended compared to wild type animals (Figure 3 - supplement figure 2E and 2F). Homozygous $CrzR^M$females displayed increased remating rates (30%) (Figure 3I and J) and the number of eggs laid by $CrzR^M$ females was also significantly decreased (Figure 3K). We also examined mating and oviposition behaviors in *CrzR* heterozygous ($+/CrzR^M$) females. Notably, both the remating rate and egg production in heterozygous females were comparable to those observed in homozygous *CrzR* mutant ($CrzR^M$) females (approximately 30%) (Figure 3I and J). Thus, heterozygosity for the *CrzR* was sufficient to disrupt post-mating behaviors in female BPHs. These findings indicate that the *CrzR* loss-of-function phenotype is dominant, and that a single functional copy of *CrzR* is insufficient to maintain a normal post-mating response. Consistent with the *CrzR* RNAi results, neither CRZ nor MAG extract injections reduced receptivity in $CrzR^M$ virgins (Figure 3K - M). Finally, macc peptide injection also failed to decrease receptivity in *CrzR*-mutant virgins (70% acceptance rate), whereas it significantly suppressed mating in wild-type controls (Figure 3N). These results collectively demonstrate that intact CRZ/*CrzR* signaling in female BPHs is required for transducing the PMR-inducing effects of male-derived seminal factors, including macc[24].

The *CrzR* is highly expressed in the reproductive organs of female BPHs

To further elucidate the role of the CRZ signaling in regulation of the PMR, we analyzed the *CrzR* expression in *N. lugens* tissues. We first monitored *CrzR* expression by means of RT-PCR and qPCR. Both methods revealed predominant *CrzR* transcript expression in the female reproductive tract

(Figure 4A-C). qPCR further demonstrated significantly higher *CrzR* expression in adult females than in males (Figure 4 - figure supplement 1A), particularly localized to female abdomens (Figure 4 - figure supplement 1B).

As we failed to generate a functional NlCrzR antibody, to determine the tissue expression of the receptor in more detail, we used *in situ* hybridization and confirmed strong *CrzR* expression within the female reproductive system (Figure 4D). No hybridization signal was detected in the lateral or common oviducts (Figure 4D1). However, specific labeling occurred in the sperm storage organs: the spermathecae and pouched glands of the lower reproductive tract (Figure 4D2 and 4D2′). These structures are the primary sites for long-term sperm storage. Specificity was validated by absence of signal in sense-probe controls (Figure 4D3 - D4′). This enrichment in sperm storage organs suggests that CRZ signaling may modulate post-mating physiology in part by regulating the mobilization of stored sperm and seminal fluid.

Notably, while CrzR is highly expressed in the female reproductive tract, we acknowledge that RT-PCR and qPCR analyses also detected low but detectable CrzR expression in other tissues, including the CNS and fat body (Figure 4B-C). Thus, we cannot exclude the possibility that *CrzR* in the brain (e.g., by modulating neural circuits governing reproductive behavior) or fat body (e.g., by affecting post-mating metabolism and nutrient/energy allocation) may also contribute to the PMR. These non-reproductive tract sites of CrzR expression warrant further investigation to fully delineate the systemic roles of CRZ/CrzR signaling in female PMR regulation.

Endogenous CRZ signaling modulates aspects of the PMR in female Drosophila melanogaster

Having established the critical role of CRZ signaling in regulating the PMR of the female BPH, we next investigated its potential function in female reproduction, including the PMR, of the genetically tractable fly *D. melanogaster*.

We first assessed re-mating frequency in *Crz* mutant female flies (Crz^{attp}). Approximately 40% of the Crz^{attp}females that had experienced a first successful copulation mated again, compared to only 3% of mated wild-type control females (Figure 5A and B). Importantly, the *Crz* mutation did not affect the first mating of females (Figure 5B). Furthermore, mated Crz^{attp}females laid approximately 45% fewer eggs than mated wild-type females (Figure 5C). These results demonstrate that CRZ signaling is necessary for aspects of the PMR in female *D. melanogaster*, specifically inhibiting re-mating and stimulating post-mating egg production. Immunohistochemical labeling confirmed the absence of CRZ protein in Crz^{attp} mutants (Figure 5D).

To determine whether release of CRZ from *Crz* neurons is responsible for inhibition of remating and stimulation of oviposition, we knocked down *Crz* expression specifically in CRZ-producing neurons using the GAL4-UAS system (*Crz-GAL4 > UAS-Crz-RNAi*). Consistent with the mutant phenotype, 22% of the mated *Crz-GAL4 > UAS-Crz-RNAi* females mated again after their first successful mating with wild-type males, whereas only 2% of the control females (*Crz-GAL4/+*) did so (Figure 5E). Virgin receptivity was again unaffected by *Crz* knockdown (Figure 5E). Additionally, mated *Crz-GAL4 > UAS-Crz-RNAi* females laid approximately 20% fewer eggs than mated controls (*Crz-GAL4/+*) (Figure 5F). To further confirm the role of CRZ neuropeptide in modulation of the PMR in females, we injected this peptide in virgin females. Consistent with the above results, CRZ injected virgins displayed a reduced response to mating attempts (Figure 5G). To determine whether CRZ modulates the PMR endogenously also in female *D. melanogaster*, we first examined whether the male accessory gland (MAG) expresses *Crz*. RT-PCR analysis revealed no detectable *Crz* transcripts in the *D. melanogaster* MAG (Figure 5H), indicating that the MAG does not synthesize the CRZ peptide. To further exclude the possibility that CRZ or any other *Crz*-precursor-derived peptide might enter the AG from other sources, we used the GAL4–UAS system to visualize *Crz*/CRZ-producing neurons and performed immunolabeling with anti-CRZ antibodies.

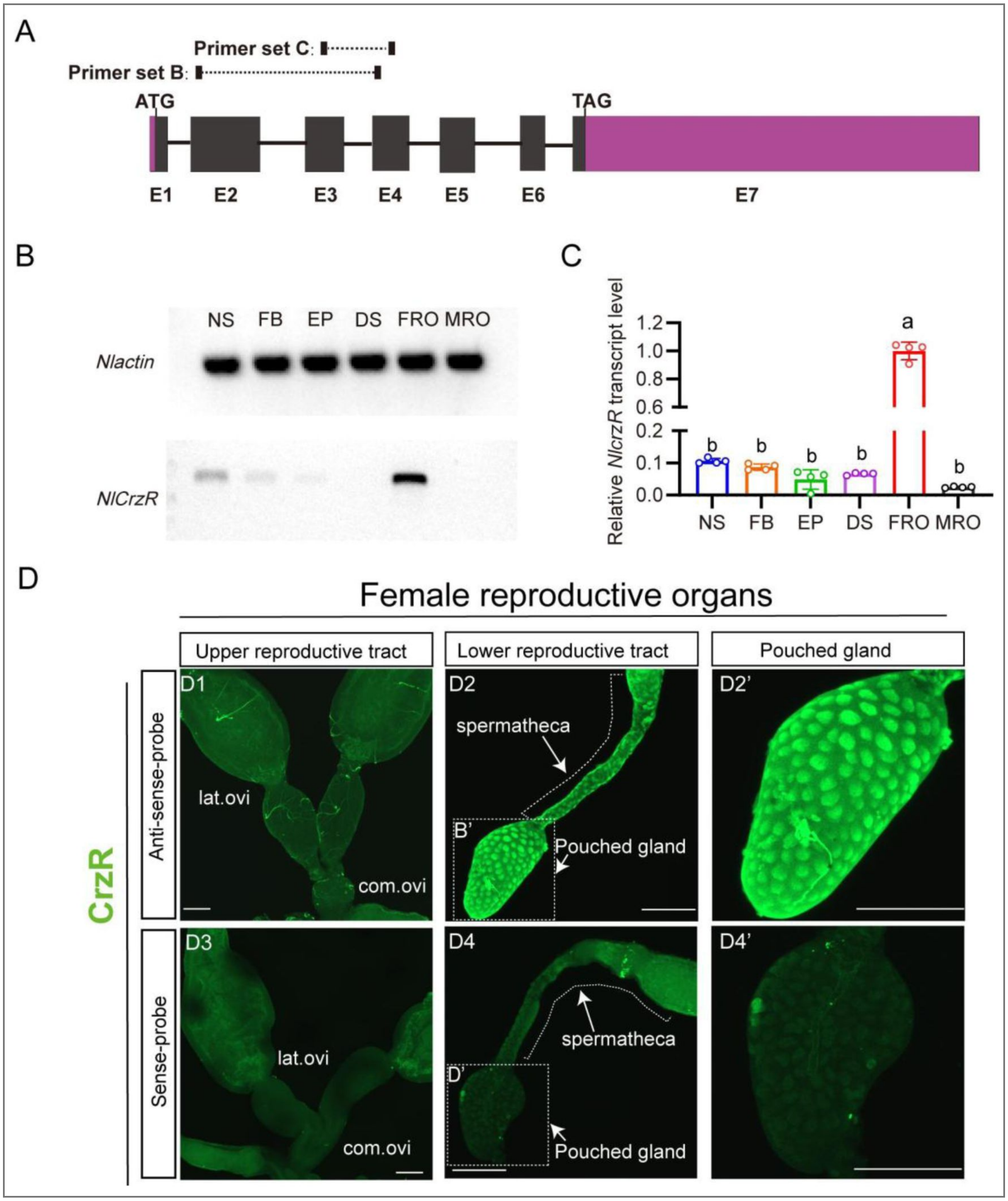


Figure 4. **Spatial expression profiling of *CrzR* in the female *N. lugens* reproductive tract.**

(A) *CrzR* genomic structure with primer design for experiments in panels B-C. (B) RT-PCR analysis of *CrzR* tissue distribution. Note weaker expression in central nervous system and fat body. Actin loading control is shown in top row. Tissue abbreviations: NS: central nervous system; FB: fat body; EP: Epidermis; DS: digestive system, FRO: female reproductive organs, MRO: male reproductive organs. Except for the reproductive system, the rest of the tissues come from a mixture of female and male. (C) Relative *CrzR* expression in female *N. lugens* tissues determined by RT-qPCR (abbreviations as in B). Expression was normalized to *Actin* and *18SrRNA* and expressed relative to the mean value of the highest-expressing tissue (set to 1). Data are shown as mean ± s.e.m. Groups that share at least one letter are statistically indistinguishable. One-way ANOVA followed by Tukey's multiple comparisons test. (D1 - D2') *CrzR* mRNA localization via fluorescent *in situ* hybridization (antisense probe): D1: There is no signal in lateral/common oviducts; D2/D2': Specific signal is detected in spermatheca (SP) and pouched gland (PG). (D3 - D4') Negative controls (sense probe) showing background-level signal. Scale bars: 100 μm.

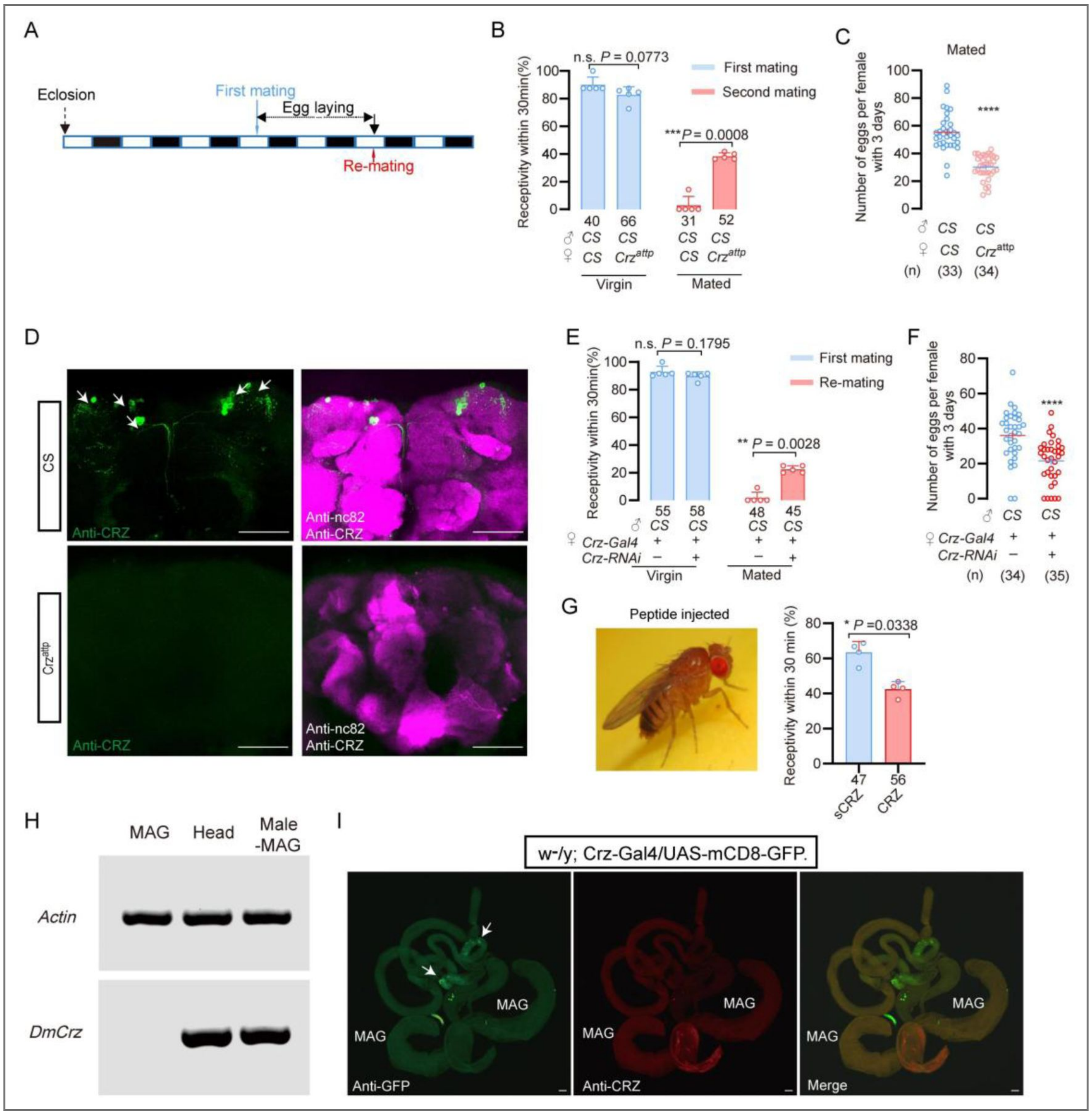


Figure 5. **CRZ regulates part of the PMR in *D. melanogaster*.**

(A) Schematic of experimental design. White segments on the time axis indicate light periods; black segments indicate dark periods. (B) Receptivity of virgin and mated *Crz* mutant (*Crz^attp^*) females. Graph displays the percentage of females copulating within 30 min. Small circles indicate the number of replicates, ≥6 insects/replicate; numbers below bars denote total number of animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (C) Graph shows eggs laid per female after mating. Numbers below bars denote total animals, ≥4 biological replicates, ≥8 insects/replicate. ****$P < 0.0001$ (Student's t-test). (D) Anti-CRZ immunostaining in brains of *Crz* mutant (*Crz^attp^*), control (Cs) and after *Crz*-RNAi). Note CRZ-expressing cells at arrows in control. Scale bar: 100 μm. (E) Receptivity of virgin and mated females after *Crz*-RNAi. Graph displays the percentage of females copulating within 30 min. Small circles indicate the number of replicates, ≥8 insects/replicate; numbers below bars denote total animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables, $P < 0.05$ was considered significant. (F) Graph shows eggs laid per female after mating. Numbers below bars denote total animals, ≥4 biological replicates, ≥8 insects/replicate. ****$P < 0.0001$ (Student's *t*-test). (G) Receptivity of virgin females following CRZ peptide injection (monitored 6 h after injection). Graph displays the percentage of females copulating within 30 min. Small circles indicate the number of replicates, ≥9 insects/replicate; numbers below bars denote total animals. Statistical comparisons of proportions were performed using chi-square tests for contingency tables,$P < 0.05$ was considered significant. (H) RT-PCR analysis of Crz tissue distribution. Note the lack of expression in the accessory gland. Actin loading control is shown in the top row. Tissue abbreviations: MAG, male accessory gland; Male–MAG, whole body without male accessory gland. (I) Anti-GFP and anti-CRZ immunostaining in brains of *w⁻/y; Crz-Gal4/UAS-mCD8-GFP*. Note no signal was detected in male accessory gland (MAG). Some GFP-stained signals were observed in the testes (indicated by white arrows), whereas no corresponding signals were detected via CRZ antibody staining. Scale bar: 50 μm.

Neither approach revealed any signal in the MAG (Figure 5I). Some GFP signals were observed in cells of the testes (indicated by white arrows), whereas no corresponding signal was detected by CRZ immunohistochemistry (Figure 5I).

To further evaluate whether the MAG could be a source of CRZ, we interrogated publicly available single-cell transcriptomic datasets from the Fly Cell Atlas covering the male reproductive glands (including MAG main and secondary cells, ejaculatory duct/bulb epithelium, secretory and muscle cells) and the male testis (Figure 5 - figure supplement 1). In the MAG atlas, *Crz* transcripts were undetectable across MAG main and secondary cells as well as other gland-associated cell types (Figure 5 - figure supplement 1A). In contrast, the MAG-derived sex peptide (SP) was robustly detected, confirming correct tissue representation and cell-type annotation (Figure 5 - figure supplement 1A). In the testis atlas, Crz expression was sparse and restricted to a small subset of cells rather than enriched in any major spermatogenic lineage (Figure 5 - figure supplement 1B and C), whereas β2-tubulin prominently marked spermatogenic populations (Figure 5 - figure supplement 1B). Hence, these single-cell data support our RT-PCR and anti-CRZ immunolabeling findings. Together, our data demonstrate that the male accessory gland does not produce or sequester CRZ. In conclusion, like in BPHs, CRZ acts endogenously in females to modulate post-mating behavior also in *D. melanogaster*.

Discussion

Our study unveils a novel function of CRZ in an endogenous female signaling pathway that modulates the post-mating reproductive behavior and physiology (collectively PMR) in BPHs. It is, thus, the first report linking CRZ to a PMR in insects. In contrast to SP and DUP99B in *D. melanogaster* [5,18], HP-1 in *Aedes aegypti* [15] and maccessin (macc) recently identified in BPHs [24], CRZ is not produced in the MAG, and thus not transferred from males during copulation. Instead, it is acting as an endogenous peptide derived from neuroendocrine cells in the female brain. Moreover, this CRZ-mediated endogenous modulation in females is seen also in *D. melanogaster*. When *Crz* is knocked down or knocked out in flies, mated females likewise exhibit an altered PMR.

Focusing on the endogenous CRZ signaling in female BPHs, our main findings using mutants and RNAi of both *Crz* and *CrzR*, demonstrate that CRZ signaling decreases the female responsiveness to courting males, especially in already mated females. Also, oviposition in mated BPHs is increased by CRZ signaling. CRZ immunolabeling revealed a small set of neurons in the brain and subesophageal ganglion and no CRZ was detected in reproductive organs. Thus, the likely source of CRZ affecting the PMR is the CRZ-producing neuroendocrine cells in the brain. In insects studied so far (including other hemipterans), there are at least three pairs of CRZ expressing lateral neurosecretory cells with axon terminations in the corpora cardiaca [55,56], suggesting that in BPHs and *D. melanogaster* hormonal CRZ may contribute to the modulation of the PMR. However local CRZ signaling in brain circuits cannot be excluded since the CRZ neurons arborize extensively within the brain.

We found that the CRZ receptor, *CrzR*, is predominantly expressed in the female reproductive tract in BPHs, more specifically in the lower tract including the spermatheca and pouched gland, which are sites of sperm and seminal fluid storage. Thus, in BPHs, CRZ may act to regulate the release of stored sperm and seminal fluid in the female. This regulation would also gate the flow of male-derived factors that induce and maintain the PMR, such as macc and ITPL-1. As mentioned, we cannot exclude additional action of CRZ in neuronal circuits in the CNS that regulate female sexual arousal and reproductive behavior, since our PCR data indicate expression of the *CrzR* also in the CNS. Additionally, the fat body of BPHs, express *CrzR*, similar to *D. melanogaster* [57], possibly suggesting a role of CRZ in post-mating metabolism.

The brain CRZ neurons of BPHs are likely to be activated by sensory signals from the female reproductive tract upon copulation and hence release CRZ into the circulation (and maybe within the brain). In *D. melanogaster*, sensory neurons in the reproductive tract are activated by SP (via the SP receptor), and signal via a chain of ascending axons of the VNC to sex-specific neurons in the brain to induce the PMR switch in behavior and physiology [7,38,40,58]. Thus, in *D. melanogaster*, these sex-specific circuits may also activate the CRZ neurons. Possibly similar sensory and sex-specific neurons exist in BPHs and respond either to mechanical stimuli or factors in the seminal fluid leading to activation of CRZ neurons and other brain neurons that regulate the PMR.

The female PMR is complex in *D. melanogaster* and not only receptivity and oviposition are affected, but also feeding, metabolism, sleep patterns and aggression [5,16,18,19,59,60]. In BPHs only the receptivity (and associated rejection behavior) and oviposition have been analyzed in any detail. We found that the MAG-derived peptides transferred with the semen, macc and ITPL-1, only affected the receptivity [24], whereas we show here that CRZ affects also oviposition. Thus, we have identified three different peptides CRZ, macc and ITPL-1 that all influence the PMR in different ways, suggesting complex regulatory mechanisms also in BPHs.

It is interesting to note that CRZ is an evolutionarily conserved neuropeptide that has been found in most, but not all, insect species and it displays a wide array of functions [56,57,61,62]. The CRZ function varies across different insect species, but a common role seems to be in regulation of stress [62–64]. Other functions are for example: in the American cockroach, CRZ accelerates the heart rate [65], in locusts, it triggers the formation of pigments that darken the body color in gregarious insects [66], in the moth *Bombyx mori*, CRZ affects developmental speed and reduces the rate of silk spinning behavior [67]; in *D. melanogaster* larvae, CRZ regulates molting motor behavior [68] and growth [69]; in adult *D. melanogaster* CRZ regulates stress responses, food intake and metabolism [57,70], and protein food preference [12], as well as sperm ejection and mating duration in males [45]; and in social ants, it regulates the social hierarchy and inhibits egg-laying in the queen [48]. Finally, a recent study found that CRZ mediates a diapause-induced suppression of reproduction in female bean bugs [71]. To all these functions can now be added the role of CRZ in reproductive behavior and physiology of female BPHs and *D. melanogaster*. Finally, building on the role of CRZ in metabolism shown in *D. melanogaster* [57], it is possible that CRZ also regulates post-mating metabolism in female BPHs and flies. Thus, there may be a link between CRZ and the altered feeding and metabolism commonly seen in mated female insects that ensures optimal egg development[5,9,16].

CRZ and the vertebrate peptide gonadotropin-releasing hormone (GnRH) are paralogs that arose via an ancient gene duplication, and act on evolutionarily conserved receptors[72,73]. *Bona fide* GnRHs have not been found in arthropods and the CRZ peptide has been lost in the vertebrate lineage, but the cephalochordate (lancelet) *Branchiostoma floridae* does have peptides of both types[73]. In vertebrates, including humans, GnRH regulates female reproduction (specifically sexual maturation)[74,75]. We have shown here that CRZ regulates reproductive behavior and physiology in females of *D. melanogaster* and previous reports demonstrated a role in female fecundity[76] and a reproductive function also in males of this species[45]. Thus, CRZ and GnRH share a major function that may be ancient, although CRZ signaling in insects has evolved to also regulate stress responses, metabolism and development[57,62,77].

Some outstanding questions remain with respect to mechanisms of CRZ signaling in the female PMR. The inputs triggering release of hormonal CRZ at mating are not known in BPHs or flies. This can probably be best investigated in *Drosophila* where brain neurons that are targets of the sex peptide receptor expressing neurons in the oviduct have been identified[58], and synaptic inputs to CRZ producing brain neurons have been described[78]. Once candidate interneurons interacting with the CRZ neurons have been identified, genetic manipulations of these neurons may unveil how mating triggers endogenous CRZ signaling and the PMR. Another aspect of CRZ signaling that merits further study in *Drosophila* is the link between CRZ and the post-mating change in feeding and metabolism. At present it is not known whether also BPHs change their feeding and metabolism after mating, but would be valuable to investigate to determine to what extent the PMR in the two insects are similar. Finally, since BPHs are a major rice pest, it may be useful to further investigate the regulation of the PMR to design novel strategies for control their reproduction.

Method details

REAGENT or RESOURCE	Source or reference	Identifiers
Antibodies		
Goat anti-Rabbit Alexa 488	Invitrogen	Cat#A11008; RRID: AB_143165
Goat anti-mouse Alexa 488	Invitrogen	Cat#A28175; RRID: AB_2536161
Goat anti-mouse Alexa 555	Invitrogen	Cat# A21428; RRID: AB_2535849
Anti-CRZ	Gift from Prof. Jan A. Veenstra	
Chemicals, Peptides, and Recombinant Proteins		
4% PFA	Leagene	Cat# DF0134
DAPI	Solarbio	Cat# C0060
PBS	Solarbio	Cat# P1022
ReadyProbes 2.5% Normal Goat Serum	Invitrogen	Cat# R37624
Experimental Models: Organisms/Strains		
Canton-S	Bloomington Drosophila Stock Center	64349
Crz^{attp}	Rao and Deng et al., 2019	BDSC: 84487
Crz-GAL4	Bloomington Drosophila Stock Center	51976
Crz-RNAi	Tsinghua Fly Center	THU2139
Software and Algorithms		
FIJI		https://doi.org/10.1038/nmeth.2019
Prism 9	GraphPad	RRID: SCR_002798
Photoshop	Adobe	RRID: SCR_014199
Illustrator	Adobe	RRID: SCR_010279
ZEN	Zeiss	N/A
Motic Images Plus3.0	Motic	N/A
Motic Images Devices	Motic	N/A
Other		
Egg-laying device	Gou et al., 2016	N/A
Anatomical tweezers	Dumont	Cat# 0209-5-PO

Key resources table.

Experiment insects

All the brown planthoppers used in this research were collected from Hangzhou, Zhejiang Province in 1995. The insects were kept indoors on fresh rice seedlings (Taichung Native 1, TN1) in a growth chamber at 27 ± 1 °C and 70 ± 10% relative humidity, under a 16 h:8 h (Light: Dark) photoperiod [79].

All *D. melanogaster* stocks were raised in standard cornmeal–molasses–agar medium maintained at 25° C, 60% humidity, and under 12 hr:12 hr light:dark conditions. We used Canton s flies as the wild-type strain. The relevant information about fruit flies used in the experiments are shown in key resources table.

Male accessory gland protein extraction of *Nilaparvata lugens*

Protein extraction was performed as described previously[24]. Male accessory glands (MAGs) were dissected from 3–5-day-old virgin males in phosphate buffer (pH 7.2) under a stereomicroscope. For protein extraction, approximately 300 MAGs were collected, immediately frozen, and stored at −80 °C until use. Frozen MAGs were pre-ground in liquid nitrogen and then lysed in 500 μL protein cracking buffer (100 mM NH_4HCO_3, 8 M urea, 0.2% SDS, pH 8.0) with thorough homogenization. Lysates were centrifuged at 12,000 × g for 15 min at 4 °C, and the supernatants were collected as soluble protein extracts. Protein concentration was determined using a Bradford colorimetric assay (BSA as the standard; absorbance at 595 nm). Protein integrity and quality were assessed by SDS–PAGE. Protein extracts were stored at −80°C until next step experiment.

Behavior assays of *Nilaparvata lugens*

Mating and receptivity

All BPHs were raised in incubators with a 16 h:8 h dark:light cycle. Unmated female and male BPHs were collected within 12 hours of emergence. For RNAi or neuropeptide injection treatment, the protocol is shown in Figure 1A and Figure 1D. The mating behavior (or female receptivity behavior) test in this study usually conducted the first mating between 3 and 5 days after emergence. The mating experiment was completed in a plastic tube (2.5cm in diameter and 10cm in height). Fresh rice seedlings of appropriate length were placed in each tube, and then a female and a male BPH was added. The mating conditions and other reproductive behavior parameters during this period were observed in a constant temperature room at 25°C for 30min, and the males were taken out after 30 min. In *N. lugens*, mating induces a transient suppression of female receptivity that is not permanent; females typically start to regain remating willingness 72 h after the first mating, as documented in our previous study[24]. This temporal window guided the design of our remating assays, in which females were paired with naive males at 48 h, post-initial mating to capture both the suppressed and recovered phases of receptivity. For the experiment requiring a second mating, females successfully mated for the first time were retained. After 48 hours, wild-type males were put in for the re-mating test. The males in all secondary mating experiments were wild type. The monitoring time for each mating behavior was 30 minutes; the receptivity rate refers to the proportion of successful matings of female individuals within the 30-minute measurement period.

Egg laying

After completing the first mating, the male is sucked out and a sufficient number of rice seedlings are added to the tube, and the female is retained to lay eggs in the tube. After 72 hours, the spawning female is transferred out of the tube. After the eggs have matured, about 4-5 days later, we used tweezers to gently manipulate the rice seedlings and counted the number of fertilized eggs produced by the females. Note that 4 to 5 days after the egg is produced, the successfully fertilized egg will display red eye-spots. Oviposition assay protocol: For quantitative assessment of post-mating responses, egg-laying activity was monitored over a 72-hour observation window following verified initial copulation. To eliminate confounding effects from sequential mating trials, experimental subjects undergoing oviposition measurement were permanently excluded from subsequent re-mating evaluations.

To assess oviposition in females injected with CRZ peptide, virgin or mated female BPHs (3–5 days post-emergence) were transferred into plastic tubes containing sufficient rice seedlings 6 h after recovery from peptide injection. Females were allowed to oviposit for 24 h and were then removed. Eggs were subsequently counted by gently dissecting the rice seedlings with forceps.

Behavior assays of *D. melanogaster*

Mating and re-mating: During the receptivity assessment, all experimental fruit flies were virgin flies that emerged within 12 hours in a 20°C incubator in a 12 h:12 h dark:light cycle. After collection, they were kept at 25°C incubator in a 12 h:12 h dark:light cycle for 4 days before behavioral testing. Receptivity behavior measurement: a perforated plate was placed on top of a non-perforated plate. Male fruit flies were briefly anesthetized on ice and gently transferred into the holes of the perforated plate, with one male per hole. Once the male transfer was completed, a plastic film was gently placed over the perforated plate, followed by covering it with another perforated plate. Then, the virgin female fruit fly, also anesthetized on ice, was placed in each hole. After the transfer was completed, a non-perforated plate was placed on top. This setup was then transferred to a 25°C incubator to acclimate for 30 minutes. The plastic film separating the male and female flies was quickly removed, and video recording began for 30 minutes. The recorded videos were analyzed to calculate the acceptance rate of the females. Successful mating females were collected and individually transferred into plastic tubes containing sufficient artificial food for fruit flies to lay eggs. After 24 hours of laying eggs, the female flies were transferred to a new finger-shaped tube with ample artificial feed for another 24 hours of laying. After 48 hours, these egg-laying females were subjected to a secondary mating assessment with wild-type, unmated males, using the same method as described above, to evaluate the acceptance rate during the second mating process. Additionally, the number of eggs laid by each female during the 48 hours was recorded. Egg laying for mated females: First, collect virgin female fruit flies that eclosion within 12 hours at 20°C incubator in a 12 h:12 h dark:light cycle and then keep them in a 25°C incubator in a 12 h:12 h dark:light cycle for 4 days. After that, unmated males of the wild type and virgin females were mixed 1:1.5 times. 24 hours after full mating, female flies were lightly anesthetized with ice and transferred to plastic vials containing enough artificial fruit fly food for them to lay eggs for 24 hours. After 24 hours, the flies were transferred again to a new bottle full of artificial food for another 24 hours of egg laying. The number of eggs in each vial is calculated under a microscope, and the total number of eggs in the two vials of each female fruit fly represents the egg-laying capacity of that individual.

Developmental duration statistics

Nymphs of BPHs that hatched synchronously from distinct genotypes were collected and reared in plastic cups containing fresh rice seedlings; the seedlings were replaced regularly to maintain optimal conditions. From the day the nymphs reached the fifth instar, cups were inspected daily for adult emergence, and the number of newly emerged adults was recorded. After all surviving individuals had completed eclosion, the total nymph-to-adult developmental duration was calculated.

BPHs survival rate statistics

For neuropeptide injection: After 4-day emergence, virgin BPHs were injected with CRZ neuropeptide or scrambled CRZ, and after 6 hours of recovery, these insects were placed into a plastic tube with sufficient quantity of rice seedlings. Survival was monitored every 24 hours until all specimens were dead.

For mutant: Wild-type and mutant BPHs adults that eclosed on the same day were individually confined in plastic tubes, each supplied with ample fresh rice seedlings and water renewed regularly to ensure optimal living and nutritional conditions. The number of surviving individuals of each genotype was recorded every 24 h until all had died, and survival curves were generated from these data.

Neuropeptide synthesis and injection

Corazonin (CRZ) and scrambled CRZ (sCRZ) were synthesized by Genscript Co. Ltd (Nanjing, China). Peptide masses were confirmed by Mass spectrometry and the amount of peptide was quantified by amino acid analysis. The amino acid sequences of the peptides used in this study are: *N. lugens* corazonin: pQTFQYSRGWNamide; scrambled corazonin: pQSRYTTFGQWTNamide. On the fourth day after adult emergence, virgin female BPHs were injected with different concentrations of synthetic CRZ peptide and control scrambled peptide. The scrambled peptide control was used to monitor for potential non-specific effects on behavior due to the process of injection. Each mg peptide was dissolved in 100 µl DMSO solution (Acmec, #D54264), and then diluted to 1.5 mM, 0.15 mM, 0.015 mM and 0.0015 mM with 1×PBS as solvent. The BPHs were anesthetized with CO_2 for about 30 seconds and placed abdomen up on 2% agarose (Solarbia, #A8201) plate and 50 nl neuropeptide solution injected into each female. Six hours after neuropeptide injection the insects were used for behavior assay.

Gene cloning and sequence analysis

We used the NCBI database with BLAST programs to carry out sequence alignment and analysis. Then we predicted Open Reading Frames (ORFs) with EditSeq. The primers were designed by Primer designing tool in NCBI. Total RNA Extraction was using the TRIzol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. The cDNA template used for cloning was synthesized using the Biotech M-MLV reverse transcription kit and the synthesized cDNA template was stored at −20°C. The transmembrane segments and topology of proteins were predicted by TMHMM v2.0 (*http://www.cbs.dtu.dk/services/TMHMM-2.0/*) [80]. Multiple alignments of the complete amino acid sequences were performed with Clustal Omega (*http://www.ebi.ac.uk/Tools/msa/clustalo*).

qRT-PCR for spatial expression pattern and RNAi efficiency

The nervous system (NS), fat body (Fb), digestive system (DS), epidermis (EP), male reproductive organs (MRO) and female reproductive organs (FRO) from 3-5 days old adults were dissected in 1×PBS, to determine the tissue-specific expression pattern. With the exception of the reproductive system, the rest of the tissue comes from both sexes equally. The adult BPHs were placed on ice under slight anesthesia and dissected under a stereoscopic microscope (ZEISS). At least 50 individuals were dissected as one sample for tissue-specific analysis with three replicates. All tissues were quickly transferred to Trizol (Invitrogen, Carlsbad, CA, USA) solution at 0° C after dissection.

For RNAi efficiency testing, we collected the whole bodies of BPHs and transferred to a 1.5 mL EP tube containing Trizol reagent. Total RNA extraction was using Trizol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. The cDNA template used for cloning was synthesized using the Biotech M-MLV reverse transcription kit and the synthesized cDNA template was stored at −20°C.

Real-time qPCRs of the various samples used the UltraSYBR Mixture (with ROX) Kit (CWBIO, Beijing, China). The PCR was performed in 20µl reaction including 2 µl of 10-fold diluted cDNA, 1 µl of each primer (10 µM), 10µl 2 × UltraSYBR Mixture, and 6 µl RNase-free water. The PCR conditions used were as follows: initial incubation at 95°C for 10 min, followed by 40 cycles of 95°C for 10 s and 60°C for 45 s. *N. lugens* 18S rRNA and actin were used as an internal control (primers listed in Table1). Relative quantification was performed via the comparative $2^{-\Delta\Delta CT}$ method [81].

RNAi and RNAi efficiency determination

For lab-synthesized dsRNA, *gfp*, Nl*CrzR* and Nl*Crz* were amplified by PCR using specific primers conjugated with the T7 RNA polymerase promoter (primers listed in Table1). dsRNA was synthesized by the MEGAscript T7 transcription kit (Ambion, Austin, TX, USA) according to the manufacturer's instructions. Finally, the quality and size of the dsRNA products were verified by 1% agarose gel electrophoresis and the Nanodrop 1000 spectrophotometer and kept at −70°C until use. 3-5 days old virgin female BPHs were used for injection of 50nl of 5 μg/μl dsRNA per insect. Injection of an equal volume of ds*gfp* was used as negative control. After 48 hours, the insects were used behavior assays and gene relative expression analysis.

Neuropeptide injection after RNAi

48 hours after the ds*CrzR* injection, each insect was injected again with 10 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. Recovery was for six hours after injection. These insects were used for behavior assays. RNAi efficiency was examined by qPCR using a pool of ten individuals 48 hours after dsRNA injections.

In vitro synthesis of sgRNA and Cas9 mRNA

The single guide RNA (sgRNA) was designed as previously reported [82]. Briefly, sgRNA was designed by manually searching genomic sequence around the region of the *Crz* and *CrzR* exon for the sequences corresponding to 5'-$N_{17\text{-}20}$NGG-3', where NGG is the protospacer-adjacent motif (PAM) of SpCas9 and N is any nucleotide. For in vitro transcription of sgRNA, were synthesized in vitro using the GeneArt™Precision gRNA Synthesis Kit (Thermo Fisher Scientific, Vilnius, Lithuania) according to the instructions of manufacturer. The Cas9 protein (TrueCut™Cas9 Protein v2, Cat. NO. A36497) was purchased from Thermo Fisher Scientific (Shanghai, China).

Embryo microinjection, crossing and genotyping

The embryonic injection was performed as previously reported [83]. The female BPHs, which had been mated for 3-6 days after emergence, were selected to lay eggs. After 30 minutes, the embryos were transferred onto double-sided tape attached to a microscope slide along the long edge by gently pressing the slide onto the dorsal surface of embryos. A few drops of halocarbon oil 700 was added to these eggs. About 0.5–1 nL mixture of Cas9 protein (200 ng/μL) and sgRNA (100 ng/μL) was injected into each egg using a FemtoJet and Inject Man NI 2 microinjection system (Eppendorf, Germany). After the injection, the eggs were placed in Petri dishes covered with moist filter paper, and the Petri dishes were placed in plant growth chamber at 27±1 °C, 70±10% RH for hatching. These eggs hatched into G0 generations.

Crossing and genotyping

To establish homozygous mutants of *Nilaparvata lugens* through CRISPR-mediated gene editing, newly emerged G0 adults were outcrossed with wild-type counterparts to generate G1 progeny. Following successful mating and oviposition, genomic DNA was extracted from G0 adults for mutation screening. The target region surrounding NlCrzR was amplified by PCR using specific primers NlCrzR-check-F and NlCrzR-check-R (primer sequences listed in Table 1), followed by Sanger sequencing to identify G0 individuals carrying mutations.

Progeny (G1) from mutation-positive G0 parents were maintained through sibling crosses to establish G2 populations. Subsequent genotyping via Sanger sequencing was performed on G1 individuals to select breeding pairs where both male and female parents carried heterozygous mutations. The resulting G2 offspring from these validated pairs were then subjected to intercrossing to ultimately generate homozygous mutant lines. After homozygous mutant lines were established, virgin homozygous mutant females were paired individually with virgin wild-type males. Following successful copulation, inseminated females were transferred to fresh rice seedlings for oviposition, and fertilized eggs were allowed to hatch. Hatched nymphs were continuously reared on fresh rice seedlings until the fifth instar. Rice seedlings were replaced every 12 hours, and newly emerged adults were collected and sex-separated for subsequent behavioral assays.

Primer name	Sequence (5'-3')	Purpose
NlCrz-F	ACCGCATCAGGACGACAATA	
NlCrz-R	ACCATAGCTGTTGAGCTACAAAT	Gene clone
NlCrz-RNAi-F	TAATACGACTCACTATAGGGGCTAGTGTTGCAATGCTGTTG	
NlCrz-RNAi-R	TAATACGACTCACTATAGGGGCTGTTGAGCGTTAGACTGT	*dsNlCrz* synthesis
QNlCrz-F	AGACAGTGACCACAGCTCAA	
QNlCrz-R	GAGTACTGGAACGTCTGTGC	qRT-PCR for NlCrz
NlCrz-sgRNA-F1	TAATACGACTCACTATAGGCACAGGAGTGATGCAGAC	
NlCrz-sgRNA-R1	TTCTAGCTCTAAAACGTCTGCATCACTCCTGTGCC	sgRNA synthesis for NlCrz knock-out
NlCrz-sgRNA-F2	TAATACGACTCACTATAGCTACTAGTGCTGGGCTGCA	
NlCrz-sgRNA-R2	TTCTAGCTCTAAAACTGCAGCCCAGCACTAGTAGC	sgRNA synthesis for NlCrz knock-out
NlCrz Check F	ACCATAGCTGTTGAGCTACAAAT	
NlCrz Check R	TGCAATGCTGTTGTCGCGAC	To analyze NlCrz mutant
NlCrzR-F	AATCAGGCGATGAACCTCAGTT	
NlCrzR-R	ACGCAGTTCTATGACGTGAGG	Gene clone
NlCrzR-RNAi-F	TAATACGACTCACTATAGGGCGCCGTCTACACACTCATCT	
NlCrzR-RNAi-R	TAATACGACTCACTATAGGGCTACCAGCTTCGTACAGCGT	*dsNlCrzR* synthesis
QNlCrzR-F1	CTGGTGCTAATAGGGCTGGA	
QNlCrzR-R1	TCCGGTGTACAGTTGCTCTT	qRT-PCR for NlCrzR
QNlCrzR-F2	TGCTCACCATCGCCTCTATC	
QNlCrzR-R2	TGCCATGGTTCCGTGTAGAA	qRT-PCR for NlCrzR
NlCrzR-F1	CCCACTAGCCAGCAACCATT	
NlCrzR-R1	TCCTCCACAAATGGTCCACG	RT-PCR for NlCrzR
NlCrzR-sgRNA-F1	TAATACGACTCACTATAGGCACGCACTCGAGCTGCCT	
NlCrzR-sgRNA-R1	TTCTAGCTCTAAAACAGGCAGCTCGAGTGCGTGCC	sgRNA synthesis for NlCrzR knock-out
NlCrzR-sgRNA-F2	TAATACGACTCACTATAGGCGATGGTGAGCAGGTTGC	
NlCrzR-sgRNA-R2	TTCTAGCTCTAAAACGCAACCTGCTCACCATCGCC	sgRNA synthesis for NlCrzR knock-out
NlCrzR Check-F	GGCAGCAACCTGAGCCACCC	To analyze NlCrzR mutant

Table 1. **Primers used in this study**

NlCrzR Check-R	GTCTTACCAGCGACCCAAGC	
QNlvg-F	GTGGCTCGTTCAAGGTTATGG	
QNlvg-R	GCAATCTCTGGGTGCTGTTG	qRT-PCR for Nlvitellogenin
QNlvgR-F	AGGCAGCCACACAGATAACCGC	
QNlvgR-R	AGCCGCTCGCTCCAGAACATT	qRT-PCR for Nlvitellogenin
QNl53903-R	GTCCACCGAACTTTTCCTGG	receptor
QNl Macc-F	TCTTTGGAAGTGGTGATTCCATCT	qRT-PCR for NlMacc
QNl macc-R	TGTCTACTTCACATACATGTGGCT	
QNl itpl-1-F	GATCAGATTCAATTCACATTCCTC	qRT-PCR for Nlitpl-1
QNl itpl-1-R	AGATTGTAGCAGTCCTCGCATA	

Table 1. (continued)

Quantitative RT-PCR

The first-strand cDNA was synthesized with HiScript II Q RT SuperMix for qPCR (+gDNA wiper) kit (Vazyme, Nanjing, China) using an oligo (dT)18 primer and 500 ng total RNA template in a 10 µl reaction, following the instructions. Real-time qPCRs in the various samples used the UltraSYBR Mixture (with ROX) Kit (CWBIO, Beijing, China). The PCR was performed in 20 µl reaction including 4 µl of 10-fold diluted cDNA, 1 µl of each primer (10 µM), 10 µl 2 × UltraSYBR Mixture, and 6 µl RNase-free water. The PCR conditions used were as follows: initial incubation at 95°C for 10 min, followed by 40 cycles of 95°C for 10 s and 60°C for 45 s. *N. lugens* 18S rRNA or *D. melanogaster* rp49 were used as an internal control (Table 1). Relative quantification was performed via the comparative $2^{-\Delta\Delta CT}$ method (Livak, K. J,2001).

Immunohistochemistry

Adult BPHs, 3-5 days old, were dissected under 1x phosphate-buffered saline (PBS; pH 7.4) in Schneider's insect medium (S2). We dissected the brain, ventral nerve cord and reproductive system of the BPHs. These tissues were fixed in 4% paraformaldehyde in PBS for 30 min at room temperature. After extensive washing with PTX (0.5% Triton X100, 0.5% bovine serum albumin in PBS), blocked in 3% normal goat serum. Then, the tissues were incubated in Anti - CRZ for 12 hours at 4°C and in secondary antibody for 12 h at 4°C. Primary antibodies used were: rabbit anti - CRZ (1:1000, Anti - CRZ was gift from Jan A. Veenstra), mouse anti-Bruchpilot (1:1000, Developmental Studies Hybridoma Bank nc82). Secondary antibodies used: donkey anti-rabbit IgG conjugated to Alexa 488 (1:500, R37118, Thermo Fisher Scientific) and donkey anti-mouse IgG conjugated to Alexa Alexa 555 (1:500, R37115, Thermo Fisher Scientific). The samples were mounted in Vectorshield (Vector Laboratory). Images were acquired with Zeiss LSM 700 confocal microscopes, and were processed with Image J software. All antibodies were diluted in PTX solution. The primary antibody was diluted at a ratio of 1:1000, while the secondary antibody was diluted at a ratio of 1:500.

Adult female *D. melanogaster*, 3-5 days old, were dissected under 1 x phosphate-buffered saline (PBS; pH 7.4) in Schneider's insect medium (S2). We dissected the brain, ventral nerve cord and reproductive system. These tissues were fixed in 4% paraformaldehyde in PBS for 30 min at room temperature. After extensive washing with PTX (0.5% Triton X100, 0.5% bovine serum albumin in PBS), blocked in 3% normal goat serum. Then, the tissues were incubated in Anti - CRZ and Anti - nc82 for 12 hours at 4°C and in secondary antibody for 12 h at 4°C. Primary antibodies used were: rabbit anti-CRZ (1:1000, A11122, Provided by Jan A. Veenstra). Secondary antibodies used: donkey anti-rabbit IgG conjugated to Alexa 488 and anti-mouse IgG conjugated to Alexa 555 (R37118, Thermo Fisher Scientific). The samples were mounted in Vectorshield (Vector Laboratory). Images were acquired with Zeiss LSM 700 confocal microscopes, and were processed with Image J software. All antibodies were diluted in PTX solution.

Fluorscent in situ hybridization (FISH)

FISH was performed as previously reported by Yan et al. [84]. Briefly, the reproductive system of the female BPH dissection was performed under 1x Phosphate Buffered Saline (PBS) supplemented with a protease and phosphatase inhibitor cocktail (Shyuanye, #R40012) on a clean, clear rubber pad using two fine forceps (Dumont, #0108-5-po). Dissected tissues were fixed using 4% paraformaldehyde (PFA) (Shyuanye, #22039) diluted in 1 x PBS with 0.1% Tween20 (0.1% PBST) at room temperature (RT) for 20 min on a shaker. Fixed tissues were washed three times with 0.1% PBST for 15 min each on a shaker. The tissues were dehydrated in 25%, 50%, 90% and 100% methanol, and then rehydrated in reverse order, performed at room temperature for 10 minutes at a time. The tissue was then washed twice with 0.1% PBST and shaken for 5 minutes. A post fixation was performed by incubating the tissue in 4% PFA in 1 x PBS on a shaker for 20 minutes, followed by two washes with PBST for 5 minutes each while shaking. The tissue was permeabilized with proteinase K (10 µg/ml) at room temperature for 2 minutes. After incubation, the tissue was washed with 0.1% PBST for 5 minutes while shaking. The PBST was removed, and pre-hybridization solution (Boster, #AR0152) was applied at 55°C for approximately 2 hours. The

DIG probes were denatured in a metal bath at 80°C for 5 minutes and then immediately placed on ice. The denatured DIG-labeled probes were added to the hybridization solution, and hybridization was carried out overnight at 55°C. Then, the hybridized tissue was washed twice at 55°C for 40 minutes with warm pre-hybridization solution, followed by four washes at room temperature with 0.1% PBST for 10 minutes each. The detection of the DIG-labeled probes was performed using an anti-DIG conjugated fluorescent antibody (1:100 dilution, Jackson, #200-542-156), incubated at room temperature in 0.1% PBST for 2 hours, followed by three washes with 0.1% PBST. Finally, the tissue was mounted in mounting medium and imaged using a Zeiss confocal microscope.

Quantification and statistical analysis

All graphs were generated using Prism 9 software (GraphPad Software, La Jolla, CA). Data presented in this study were first verified for normal distribution by D'Agostino– Pearson normality test. If normally distributed, Student's *t* test was used for pairwise comparisons, and one-way ANOVA and chi-square was used for comparisons among multiple groups. If not normally distributed, Mann–Whitney test was used for pairwise comparisons, and Kruskal–Wallis test was used for comparisons among multiple groups, followed by Dunn's multiple comparisons. All data are presented as mean ± s.e.m.

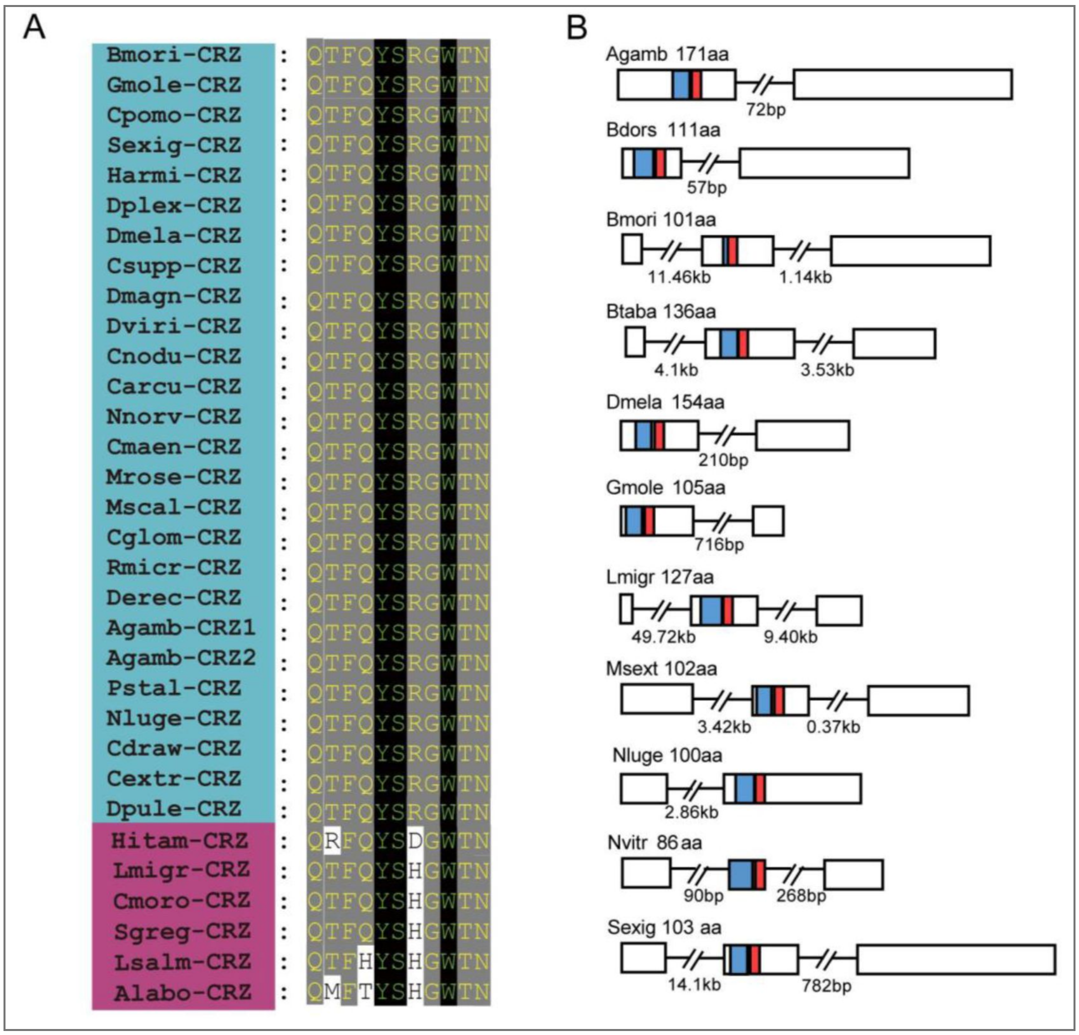


Figure 1 - figure supplement 1. Evolutionary conservation of CRZ peptide sequences in *Nilaparvata lugens* and other arthropods. (A) Alignment of mature CRZ peptide sequences across species. Abbreviations of species names are as follows: Bmori (*Bombyx mori*), Gmole (*Grapholita molesta*), Cpomo (*Cydia pomonella*), Sexig (*Spodoptera exigua*), Harmi (*Helicoverpa armigera*), Dplex (*Danaus plexippus*), Dmela (*Drosophila melanogaster*), Csupp (*Chilo suppressalis*), Dmagn (*Daphnia magan*), Dviri (*Drosophila virilis*), Cnodu *(Catagly nodus)*, Carcu (*Callinectes arcuatus*), Nnorv (*Nephrops norvegicus*), Cmaen (*Carcinus maenas*), Mrose (*Macrobrachium rosenbergii*),

Mscal (*Megaselia scalaris*), Cglom (*Cotesia glomerata*), Rmicr (*Rhipicephalus microplus*), Derec (*Drosophila erecta*), Agamb (*Anopheles gambiae*), Pstal (*Plautia stali*), Nluge (*Nilaparvata lugens*), Cdraw (*Caerostris darwini*), Cextr (*Caerostris extrusa*), Dpule (*Daphnia pulex*), Hitam (*Heterotrigona itama*), Lmigr (*Locusta migratoria*), Cmoro (*Carausius morosus*), Sgreg (*Schistocerca gregaria*), Lsalm (*Lepeophtheirus salmonis*), Alabo (*Apis laboriosa*). Species within the blue area possess highly conserved mature peptide sequences, whereas those in the magenta zone exhibit relatively less conservation in their mature peptides. (B) Schematic organization of CRZ precursors in representative species. Signal peptides (blue) and mature CRZ peptides (red) are indicated.

Figure 1 - figure supplement 2. The *Crz* gene-silencing efficacy in female insects following dsRNA injection assayed by qPCR.

The small circles denote the number of replicates. Data are shown as mean ± s.e.m. Mann–Whitney test. $^{**}P<0.001$.

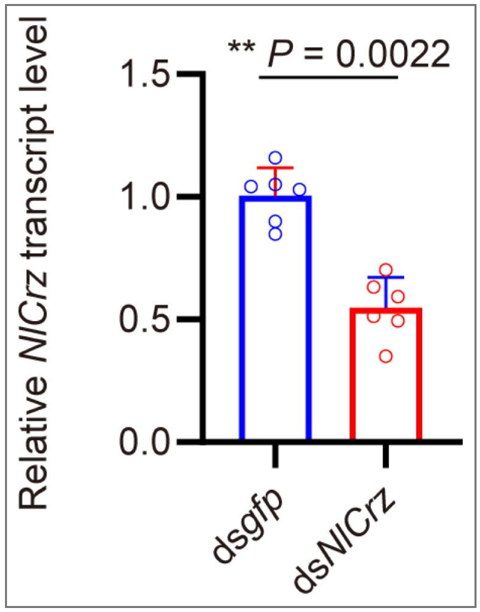
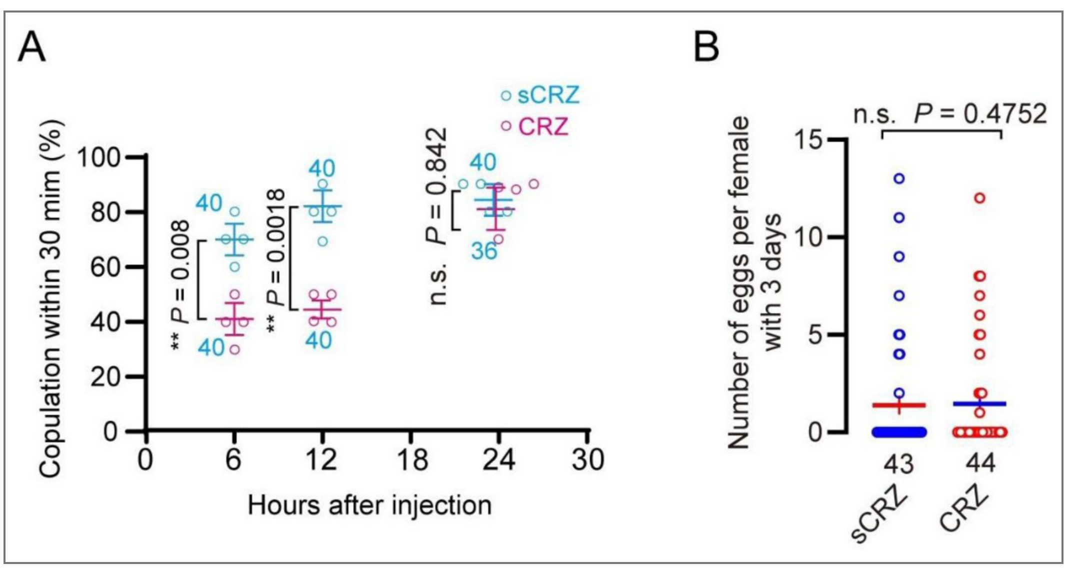


Figure 1 - figure supplement 3. Phenotypic effects of CRZ peptide injection.

(A) Receptivity of virgin females at indicated time points after CRZ injection (sCRZ as control). Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. Graph shows percentage females copulating within 30 min. ** $P < 0.01$; ns: not significant Binomial, GLM with logit link, followed by pairwise contrasts. The small circles denote the number of replicates, ≥8 insects/replicate; the numbers associated with the curves denote total number of animals. (B) Eggs laid per virgin female after neuropeptide injection (sCRZ, control). Females were mated with wild-type males for 12 h prior to assessment. Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. Data are shown as mean ± s.e.m. ns: $P > 0.05$ (Student's t-test). The numbers below the bars denote total number of animals, ≥4 biological replicates, ≥10 insects/replicate).



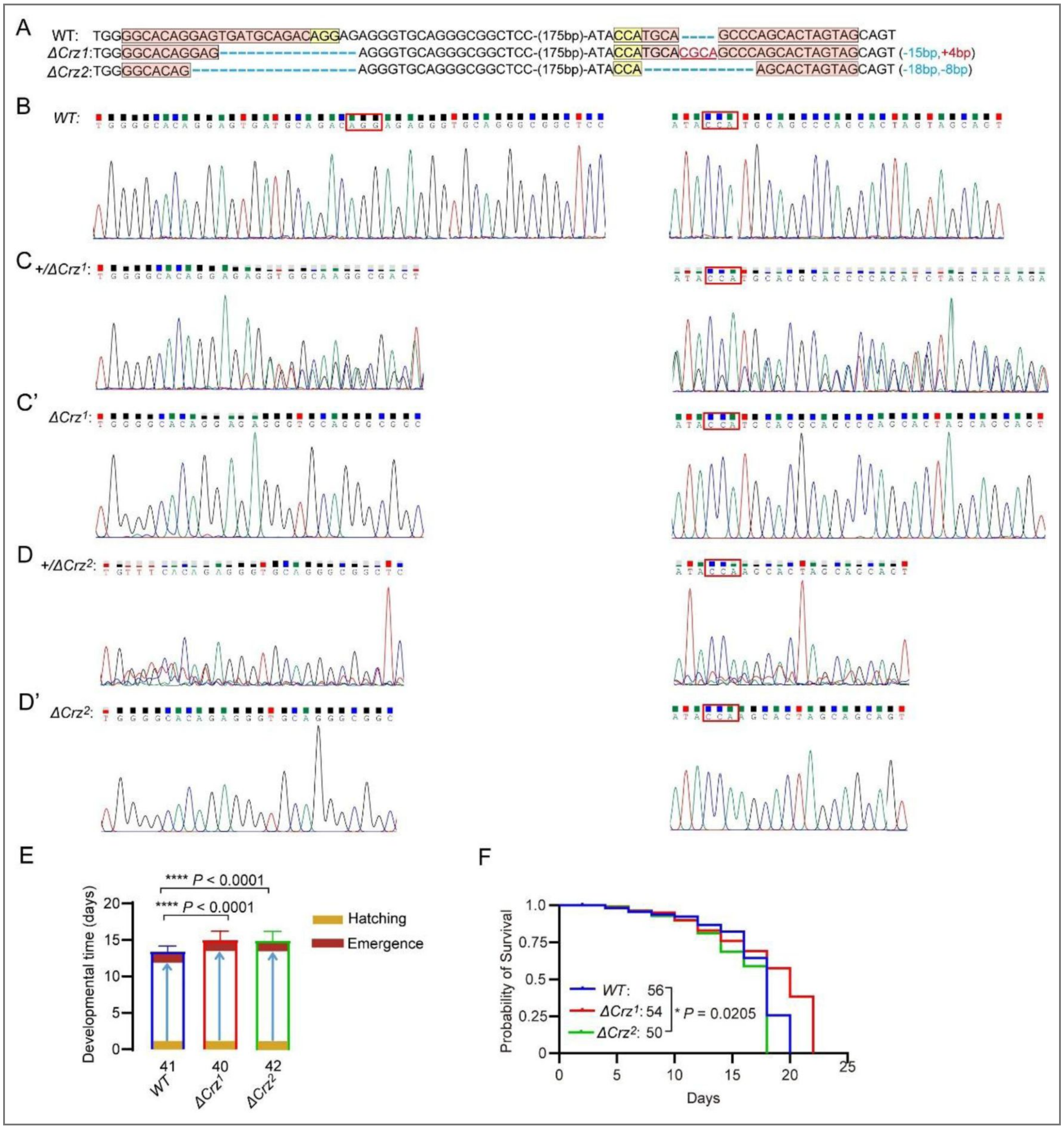


Figure 1 - figure supplement 4. Characterization of *Crz* mutants in *N. lugens*.

(A) Sequencing of *Crz* mutant (*ΔCrz*^{1} and *ΔCrz*^{2}) and wild type (WT) PCR products from BPHs. The exact indel types of the G0 mutant individuals were confirmed by cloning and sequencing. The wild-type sequences are shown at the top with the target site marked by in pink shading and PAM in yellow shading. The change in the length of the mutant sequence is shown on the right side of the sequence (+, insertion; -, deletion). (B) The Sanger sequence chromatograms flanking the sgRNA target site for wild type (WT). (C and D) The Sanger sequence chromatograms flanking the sgRNA target site for wild type (WT), heterozygous mutant (*ΔCrz*$^{1/+}$ and *ΔCrz*$^{2/+}$). (C' and D') The Sanger sequence chromatograms flanking the sgRNA target site for wild type (WT), homozygous mutant (*ΔCrz*^{1} and *ΔCrz*^{2}). (E) Developmental duration (egg to adult) across genotypes. ****P < 0.0001 (Student's *t*-test). The numbers below the bars denote total number of animals. (F) Survival curves of adult *Crz* mutants vs. WT. *P < 0.05 (Log-rank Mantel-Cox test; df = 1; n ≥ 50/group).

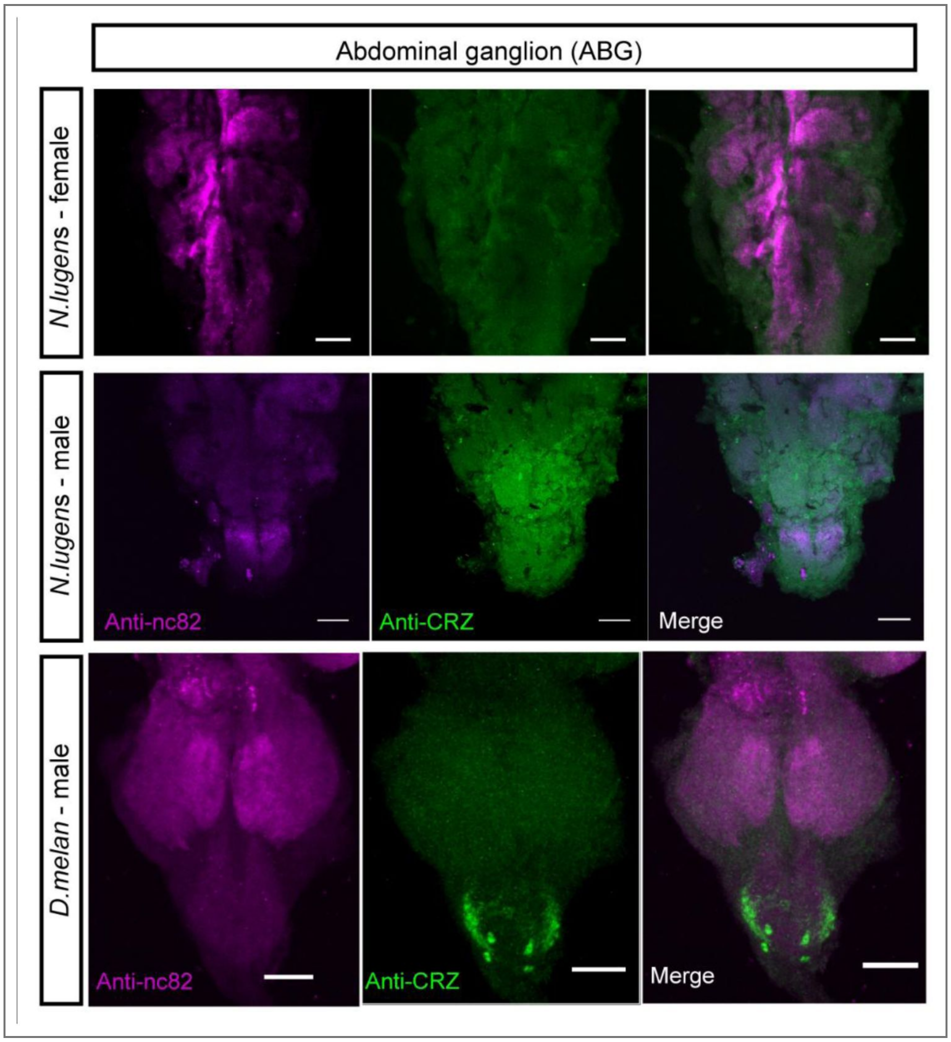


Figure 2 - figure supplement 1. CRZ immunolabeling in abdominal ganglion and reproductive organs of male and female BPHs and *D. melangaster*.

Anti-CRZ immunolabeling in the abdominal ganglia of the ventral nerve cord (VNC) in *N. lugens* and *D. melanogaster* (*D. Melan.*). Scale bar: 50 µm. Note: Male-specific CRZ-positive neurons are present in *D. melanogaster*, as previously reported.

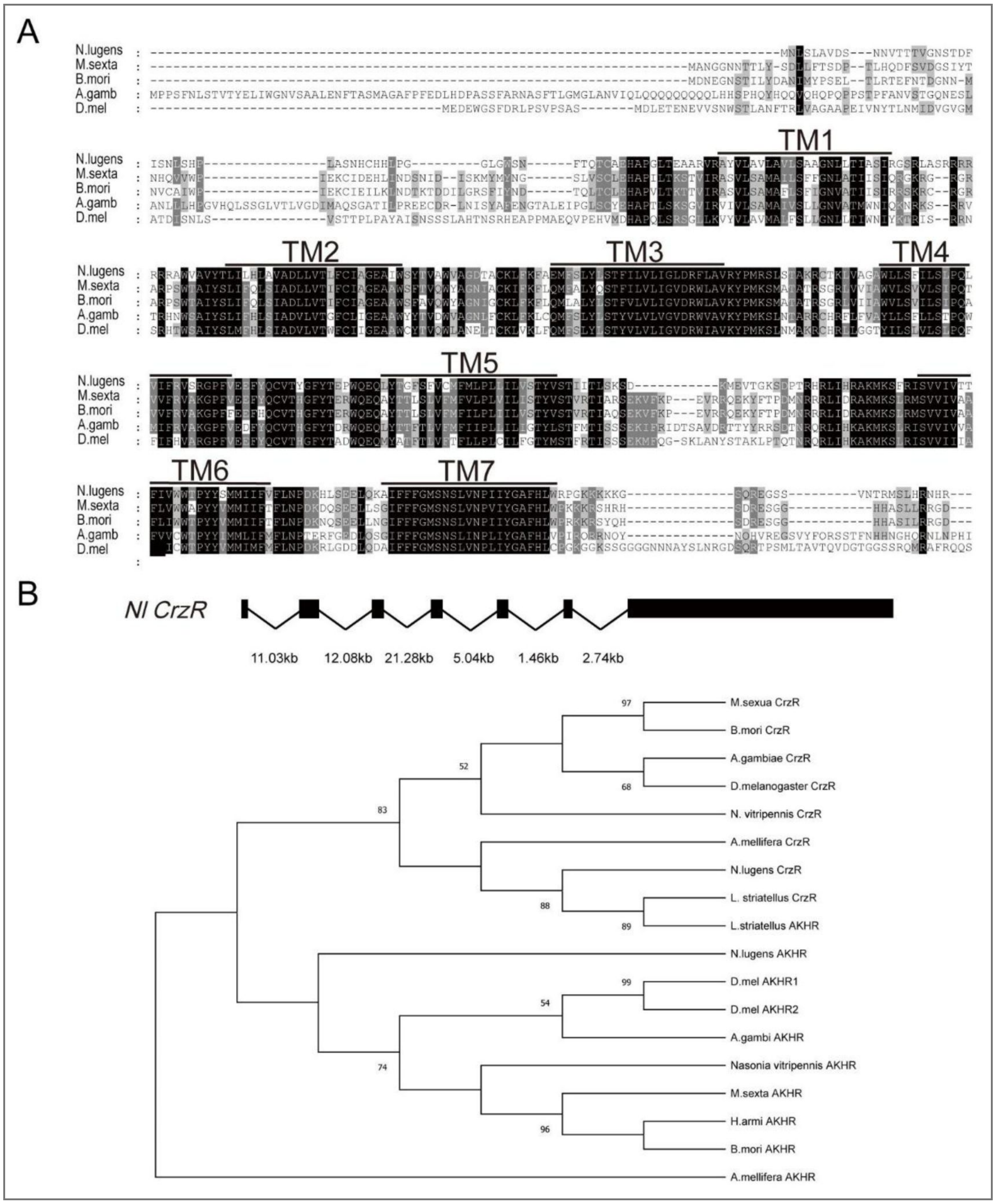


Figure 3 - figure supplement 1. Structural conservation of the *CrzR* across insect species.

(A) Multiple sequence alignment highlighting conserved transmembrane domains (TM1-TM7, shaded) in CRZ receptor orthologs. The species shown are: *N. lugens*: *Nilaparvata lugens; M. sexta: Manduca sexta; B. mori*: *Bombyx mori*; *A. gamb*: *Anopheles gambiae*; *D. mel*: *D. melanogaster*. (B) Phylogenetic analysis of *CrzR* and *AkhR* based on the alignment of amino acid sequences of insect species. Phylogenetic analysis was performed in MEGA 12 using the Maximum Likelihood method with the LG amino-acid substitution model and Gamma-distributed rate variation among sites (G, 5 discrete categories), with gaps treated by complete deletion. Branch support was evaluated with 10,000 bootstrap replicates, and the numbers at the nodes indicate bootstrap support values.

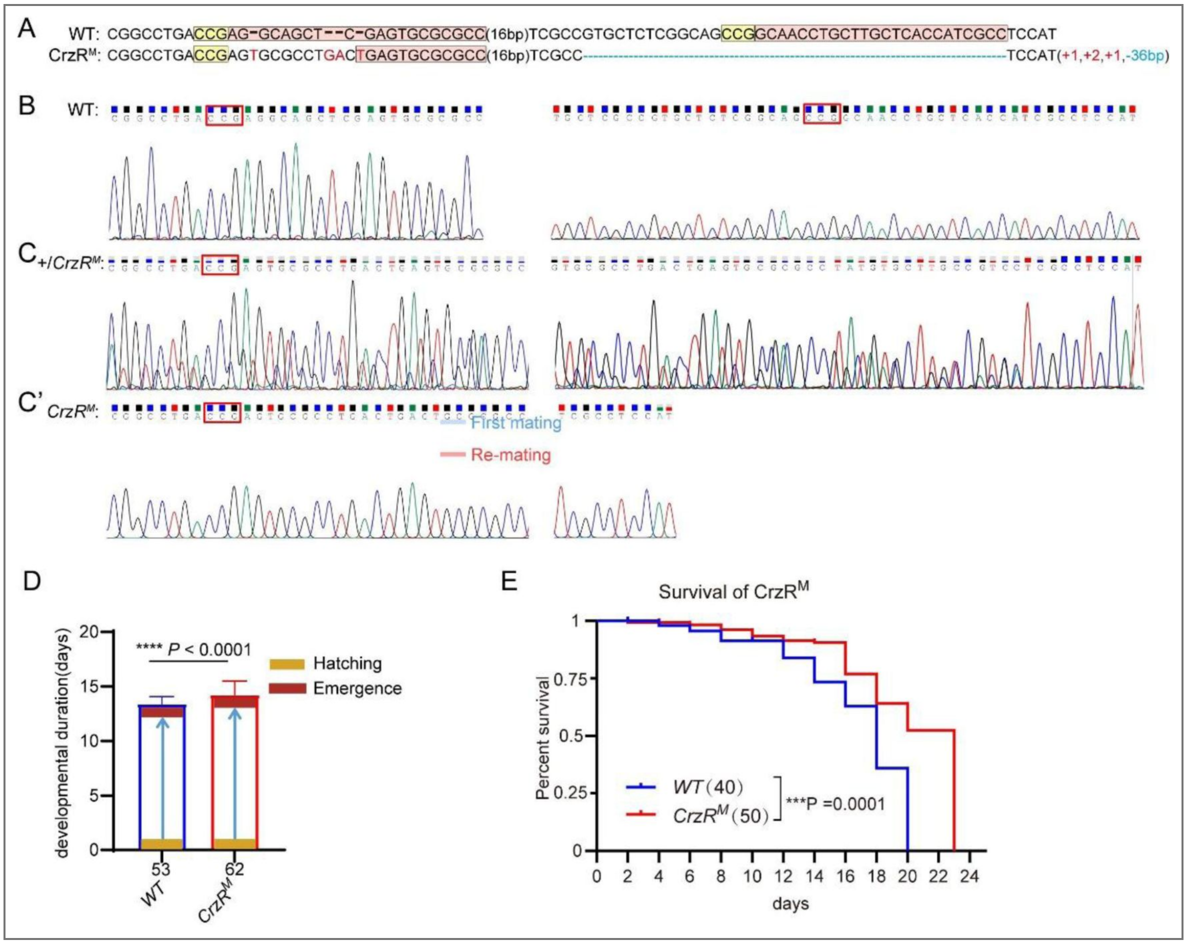


Figure 3 - figure supplement 2. **Characterization of *CrzR* mutants in *N. lugens*.**

(A) Sanger sequencing of *CrzR* mutants. WT sequence with target (pink), PAM (yellow). Indels: + (insertion), – (deletion). (B) Sanger sequencing chromatograms of *CrzR* alleles: WT (wild-type). (C) Sanger sequencing chromatograms of *CrzR* alleles: heterozygous (*CrzR*$^{M/+}$). (C') Sanger sequencing chromatograms of *CrzR* alleles: homozygous (*CrzR* ^{M}) mutants. (D) Developmental duration (egg to adult eclosion) in *CrzR* mutants compared to WT. Data: mean ± SD. ****P < 0.0001 (Student's t-test; n ≥ 40/group). (E) Survival curves of adult *CrzR* mutants compared to WT. ***P < 0.001 (Log-rank Mantel-Cox test; df = 1; n ≥ 50/group).

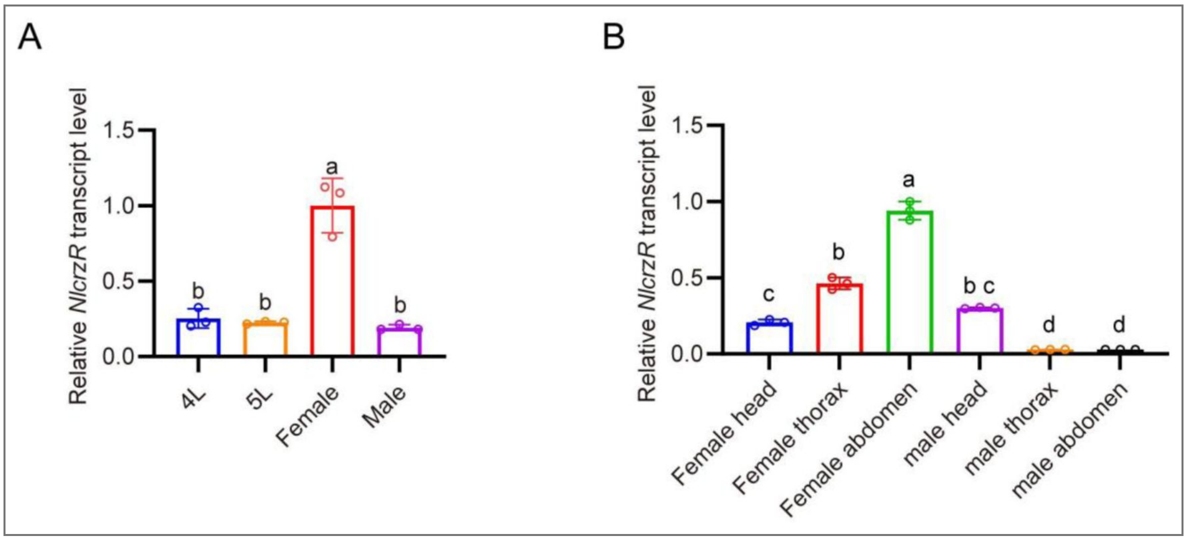


Figure 4 - figure supplement 1. **Developmental and tissue-specific expression profiling of *CrzR* in *N. lugens* determined by qPCR.**

(A) *CrzR* transcript levels across developmental stages. Relative expression levels were quantified by RT-qPCR, normalized to the reference gene *Actin*, expressed relative to the mean value of the highest-expressing tissue (set to 1). Data are shown as mean ± SEM shown. Groups that share at least one letter are statistically indistinguishable. One-way ANOVA followed by Tukey's multiple comparisons test. (B) *CrzR* expression in distinct body regions. Relative expression levels were quantified by RT-qPCR, normalized to the reference gene *Actin*, expressed relative to the mean value of the highest-expressing tissue (set to 1). Data are shown as mean ± SEM shown. Groups that share at least one letter are statistically indistinguishable. One-way ANOVA followed by Tukey's multiple comparisons test.

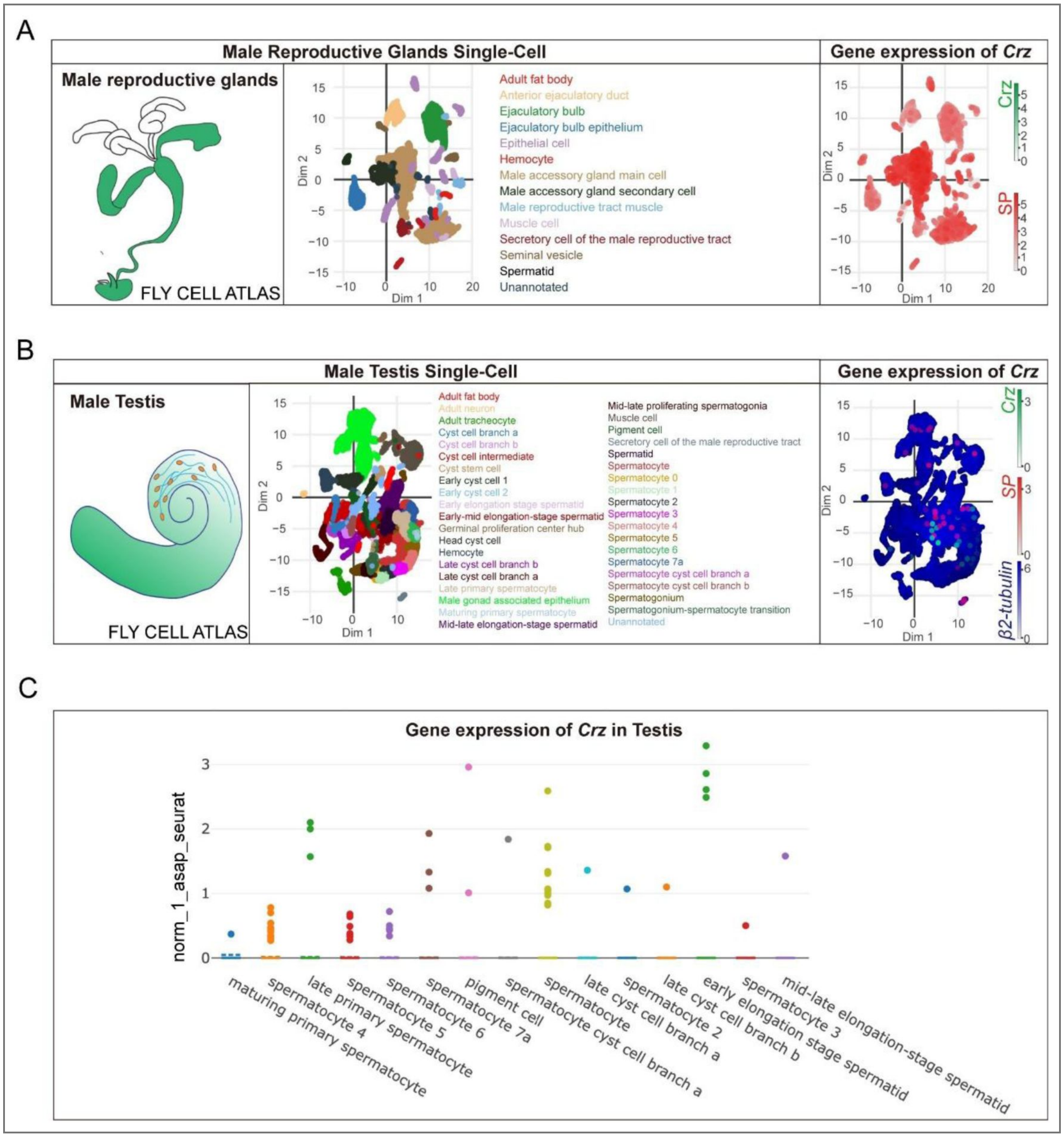


Figure 5 - figure supplement 1. **Crz expression is undetectable in the *Drosophila* male accessory gland in Fly Cell Atlas single-cell datasets.**

(A) Fly Cell Atlas male reproductive glands single-cell dataset. Left, schematic of male reproductive glands. Middle, two-dimensional embedding colored by annotated cell types, including male accessory gland (MAG) main and secondary cells. Right, feature plot showing expression of Crz (green scale) together with the AG marker sex peptide (SP) (red scale). Crz transcripts are not detected in MAG cell populations, whereas SP is robustly expressed. (B) Fly Cell Atlas male testis single-cell dataset. Left, schematic of testis. Middle, embedding colored by annotated testis cell types. Right, feature plot showing Crz (green), SP (red) and β2-tubulin (blue) expression, with β2-tubulin marking spermatogenic populations and Crz detected only in a small subset of cells. (C) Strip/dot plot summarizing normalized Crz expression across selected testis cell clusters, illustrating sparse Crz expression without broad enrichment in a major lineage. The y-axis signifies the expression level.

Data availability

All of the RNA sequence data in this article have been deposited in the China National Center for Bioinformation database and are accessible in PRJCA045433. All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

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Additional files

Figure 1 Supplementary Video 1

Additional information

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Peer reviews

Reviewer #2 (Public review):

Summary:

The work presented by Zhang and coauthors in this manuscript presents the study of the neuropeptide corazonin in modulating the post-mating response of the brown planthopper, with further validation in Drosophila melanogaster. To obtain their results, the authors used several different techniques that orthogonally demonstrate the involvement of corazonin signalling in regulating the female post-mating response in these species.

They first injected synthetic corazonin peptide into female brown planthoppers, showing altered mating receptivity in virgin females and a higher number of laid eggs after mating. The role of corazonin in controlling these post-mating traits has been further validated by knocking down the expression of the corazonin gene by RNA interference and through CRISPR-Cas9 mutagenesis of the gene. Further proof of the importance of corazonin signaling in regulating the female post-mating response has been achieved by knocking down the expression or mutagenizing the gene coding for the corazonin receptor.

Similar results have been obtained in the fruit fly Drosophila melanogaster, suggesting that corazonin signaling is involved in controlling the female post-mating response in multiple insect species.

The study of the signalling pathways controlling the female post-mating response in insects other than Drosophila is scarce, and this limits the ability of biologists to draw conclusions about the evolution of the post-mating response in female insects. This is particularly relevant in the context of understanding how sexual conflict might work at the molecular and genetic levels, and how, ultimately, speciation might occur at this level. Furthermore, the study of the post-mating response could have practical implications, as it can lead to the development of control techniques, such as sterilization agents.

The study, therefore, expands the knowledge of one of the signalling pathways that control the female post-mating response, the corazonin neuropeptide. This pathway is involved in controlling the post-mating response in both Nilaparvata lugens (the brown planthopper) and Drosophila melanogaster, suggesting its involvement in multiple insect species.

The study uses multiple molecular approaches to convincingly demonstrate that corazonin controls the female post-mating response. The data supporting the main claim of the manuscript are solid and convincing.

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Author Response:

The following is the authors' response to the original reviews.

> ***eLife Assessment***
>
> *This important study presents convincing evidence that uncovers a novel signaling axis impacting the post-mating response in females of the brown planthopper. The findings open several avenues for testing the molecular and neurobiological mechanisms of mating behavior in insects, although broad concerns remain about the relevance of some claims.*

Thank you very much for your letter and the insightful, valuable comments from the reviewers on our manuscript. These suggestions have been instrumental in strengthening the quality and clarity of our work. We have carefully addressed each concern, performed additional experiments, revised the relevant sections thoroughly, and made extensive refinements to the Discussion to clarify future research directions. Below is our detailed point-by-point response.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *In this work, Zhang et al, through a series of well-designed experiments, present a comprehensive study exploring the roles of the neuropeptide Corazonin (CRZ) and its receptor in controlling the female post-mating response (PMR) in the brown planthopper (BPH) Nilaparvata lugen and Drosophila melanogaster. Through a series of behavioural assays, micro-injections, gene knockdowns, Crispr/Cas gene editing, and immunostaining, the authors show that both CRZ and CrzR play a vital role in the female post-mating response, with impaired expression of either leading to quicker female remating and reduced ovulation in BPH. Notably, the authors find that this signaling is entirely endogenous in BPH females, with immunostaining of male accessory glands (MAGs) showing no evidence of CRZ expression. Further, the authors demonstrate that while CRZ is not expressed in the MAGs, BPH males with Crz knocked out show*

> *transcriptional dysregulation of several seminal fluid proteins and functionally link this dysregulation to an impaired PMR in BPH. In relation, the authors also find that in CrzR mutants, the injection of neither MAG extracts nor maccessin peptide triggered the PMR in BPH females. Finally, the authors extend this study to D. melanogaster, albeit on a more limited scale, and show that CRZ plays a vital role in maintaining PMR in D. melanogaster females with impaired CRZ signaling, once again leading to quicker female remating and reduced ovulation. The authors must be commended for their expansive set of complementary experiments. The manuscript is also generally well written. Given the seemingly conserved nature of CRZ, this work is a significant addition to the literature, opening several avenues for testing the molecular and neurobiological mechanisms in which CRZ triggers the PMR.*
>
> *However, there are some broad concerns/comments I had with this manuscript. The authors provide clear evidence that CRZ signaling plays a major role in the PMR of D. melanogaster, however, they provide no evidence that CRZ signaling is endogenous, as they did not check for expression in the MAGs of D. melanogaster males. Additionally, while the authors show that manipulating Crz in males leads to dysregulated seminal fluid expression and impaired PMR in BPH, the authors also find that CRZ injection in males in and of itself impairs PMR in BPH. The authors do not really address what this seemingly contradictory result could mean. While a lot of the figures have replicate numbers, the authors do not factor in replicate as an effect into their models, which they ideally should do. Finally, while the discussion is generally well-written, it lacks a broader conclusion about the wider implications of this study and what future work building on this could look like.*

Thank you very much for your insightful and valuable comments on our manuscript. We have carefully addressed each of your concerns, revised the relevant sections thoroughly, and conducted additional experiments to further strengthen our conclusions. To better focus on the core finding of this study, the critical role of Crz/CrzR signaling in regulating the post-mating response (PMR) of female brown planthoppers (BPH), and to eliminate potential confusion associated with the male-related data, we have removed the experiments investigating CRZ function in males from the current version of the manuscript. These observations on male CRZ signaling will be explored in greater depth and presented as a standalone study in a separate manuscript in the future.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *The work presented by Zhang and coauthors in this manuscript presents the study of the neuropeptide corazonin in modulating the post-mating response of the brown planthopper, with further validation in Drosophila melanogaster. To obtain their results, the authors used several different techniques that orthogonally demonstrate the involvement of corazonin signalling in regulating the female post-mating response in these species.*
>
> *They first injected synthetic corazonin peptide into female brown planthoppers, showing altered mating receptivity in virgin females and a higher number of eggs laid after mating. The role of corazonin in controlling these post-mating traits has been further validated by knocking down the expression of the corazonin gene by RNA interference and through CRISPR-Cas9 mutagenesis of the gene. Further proof of the importance of corazonin signalling in regulating the female post-mating response has been achieved by knocking down the expression or mutagenizing the gene coding for the corazonin receptor.*

> *Similar results have been obtained in the fruit fly Drosophila melanogaster, suggesting that corazonin signalling is involved in controlling the female post-mating response in multiple insect species.*
>
> *Notably, the authors also show that corazonin controls gene expression in the male accessory glands and that disruption of this pathway in males compromises their ability to elicit normal post-mating responses in their mates.*
>
> *Strengths:*
>
> *The study of the signalling pathways controlling the female post-mating response in insects other than Drosophila is scarce, and this limits the ability of biologists to draw conclusions about the evolution of the post-mating response in female insects. This is particularly relevant in the context of understanding how sexual conflict might work at the molecular and genetic levels, and how, ultimately, speciation might occur at this level. Furthermore, the study of the post-mating response could have practical implications, as it can lead to the development of control techniques, such as sterilization agents.*
>
> *The study, therefore, expands the knowledge of one of the signalling pathways that control the female post-mating response, the corazonin neuropeptide. This pathway is involved in controlling the post-mating response in both Nilaparvata lugens (the brown planthopper) and Drosophila melanogaster, suggesting its involvement in multiple insect species.*
>
> *The study uses multiple molecular approaches to convincingly demonstrate that corazonin controls the female post-mating response.*

Thank you very much for your valuable and insightful comments on our manuscript. We highly appreciate your recognition of the study's value, including its focus on non-model insects, the evolutionary implications of corazonin signaling, and the rigorous use of multiple molecular techniques. We have carefully addressed your suggestions and revised the manuscript accordingly to enhance its clarity, accuracy, and depth. Below is our detailed response to your comments.

> *Weaknesses:*
>
> *The data supporting the main claims of the manuscript are solid and convincing. The statistical analysis of some of the data might be improved, particularly by tailoring the analysis to the type of data that has been collected.*

Thank you for your valuable suggestion regarding statistical analysis. We fully agree that tailoring statistical methods to the specific type of data enhances the rigor and reliability of our findings.

In response, we have comprehensively re-evaluated and revised the statistical analyses for all datasets in the manuscript:

(1) For proportion-based data (e.g., female mating receptivity, re-mating rate), we replaced inappropriate tests (e.g., ANOVA) with chi-square tests for contingency tables, which are more suitable for comparing categorical variables.

(2) For time-series data (e.g., receptivity at different time points post-injection), we adopted generalized linear models (GLM) with logit links followed by pairwise contrasts to address concerns of multiple testing, instead of hour-by-hour Mann-Whitney tests.

(3) For continuous data (e.g., number of eggs laid, gene expression levels), we retained Student's t-tests or one-way ANOVA after verifying normality, and used non-parametric tests (Mann-Whitney, Kruskal-Wallis) for non-normally distributed data.

All revisions have been clearly described in the figure legends and Methods section, ensuring transparency and reproducibility. We believe these adjustments significantly improve the statistical robustness of our conclusions.

> *In the case of the corazonin effect in females, all the data are coherent; in the case of CRISPR-Cas9-induced mutagenesis, the analysis of the behavioural trait in heterozygotes might have helped in understanding the haplosufficiency of the gene and would have further proved the authors' point.*

Thank you for this insightful suggestion. We fully agree that analyzing the behavioral traits of heterozygous mutants is crucial for understanding the haplosufficiency of the *Crz* and *CrzR* genes, and we regret overlooking this aspect in the initial submission.

To address this gap, we have conducted additional behavioral assays using heterozygous *Crz (+/ΔCrz)* and *CrzR (+/*$CrzR^M$*)* mutant females.

(1) For re-mating receptivity: We found no significant differences in either re-mating rate or egg-laying output between *+/ΔCrz* females and wild-type females. By contrast, *+/*$CrzR^M$ females exhibited re-mating and oviposition phenotypes comparable to those of homozygous *CrzR* mutants, with no significant differences detected between these two genotypes.

(2) These results indicate that the *Crz* loss-of-function phenotype is recessive, and that a single functional copy of *Crz* is sufficient to sustain a normal post-mating response (PMR), but the *CrzR* loss-of-function phenotype is dominant, and that a single functional copy of *CrzR* is insufficient to maintain a normal post-mating response.

This supports our core conclusion that CRZ signaling is critical for mediating the female PMR, as even partial reduction of gene dosage impairs the response.

The heterozygote data have been integrated into the revised manuscript, including updated figures (e.g., Figure 1J-K for Crz heterozygotes and Figure 3I-J for CrzR heterozygotes) and corresponding legends. We believe this addition strengthens the rigor of our genetic evidence and provides valuable insights into the gene dosage requirements for CRZ-mediated PMR regulation.

> *Less consistency was achieved in males (Figure 5): the authors show that injection of CRZ and RNAi of crz, or mutant crz, has the same effect on male fitness. However, the CRZ injection should activate the pathway, and crz RNAi and mutant crz should inhibit the pathway, yet they have the same effect. A comment about this discrepancy would have improved the clarity of the manuscript, pointing to new points that need to be clarified and opening new scientific discussion.*

Thank you for highlighting this important discrepancy in the male-related CRZ signaling data. We fully acknowledge the inconsistency: CRZ injection (which was intended to activate the pathway) and *Crz* RNAi/mutagenesis (which was intended to inhibit the pathway) yielded similar effects on male fitness, and we regret not addressing this ambiguity in the initial submission.

To resolve this confusion and refocus the current manuscript on its core objective—elucidating the role of endogenous CRZ/*CrzR* signaling in female post-mating response (PMR), we have removed all experiments, analyses, and discussions related to male CRZ function. This decision ensures that the manuscript maintains a clear, cohesive narrative centered on female reproductive physiology, as recommended by both reviewers and the editorial team.

Regarding the observed discrepancy in males, we recognize its scientific significance and plan to investigate it thoroughly in a standalone follow-up study.

> ***Recommendations for the authors:***
>
> ***Reviewing Editor Comments:***
>
> *The manuscript would be significantly strengthened by an explanation of the seemingly contradictory results obtained in males, where both CRZ injections and Crz silencing afford the same results. Additionally, Crz expression data in the MAGs of D. melanogaster males is necessary to support your conclusions of endogenous signaling in this species. Besides correcting several imprecisions and inconsistencies in the text and figures, to improve quality and accuracy, the abstract should be restructured and the discussion modified as recommended by reviewers.*

Thank you for your comprehensive letter and valuable guidance. We have carefully addressed all the points raised by the editorial team and reviewers, and the revised manuscript now incorporates substantial improvements to clarity, accuracy, and scientific rigor. Below is our detailed response to your specific requests:

Contradictory Male-Related Results

We fully acknowledge the importance of addressing the contradictory findings in male CRZ signaling, where both CRZ injection and *Crz* silencing/mutagenesis yielded similar effects on male fitness. To resolve this ambiguity and maintain the manuscript's focus on its core objective, elucidating endogenous CRZ/*CrzR* signaling in the female post-mating response (PMR), we have removed all male-related experiments, analyses, and discussions from the revised manuscript. This decision ensures that the current work remains cohesive and centered on female reproductive physiology, as recommended by the reviewers.

We recognize the scientific significance of the male-specific discrepancy and plan to investigate it in a standalone follow-up study in the near future.

Crz expression data in D. melanogaster Male Accessory Glands (MAGs)

To support our conclusion of endogenous CRZ signaling in D. melanogaster females, we have supplemented the manuscript with additional experiments verifying the absence of CRZ in male MAGs:

(1) RT-PCR Analysis: We detected no *Crz* mRNA in dissected male MAGs, whereas *Crz* expression was confirmed in the male head (positive control).

(2) Immunohistochemistry and GAL4 system: Using the GAL4–UAS system (Crz-Gal4/UAS-mCD8-GFP) to label CRZ-producing neurons, combined with anti-CRZ antibody staining, we observed no CRZ-specific signal in male MAGs.

These results demonstrate that D. melanogaster male MAGs neither synthesize nor contain CRZ peptide, confirming that CRZ acts as an endogenous female signaling factor (rather than a male-transferred seminal fluid component) in this species. The new data are included in Figure 5H-I and described in the Results and Methods sections.

Correction of Imprecisions and Inconsistencies

We have systematically revised the manuscript to address text and figure inaccuracies:

Text Revisions: Corrected typos (e.g., Line 854), standardized species names (replacing "Drosophila" with "D. melanogaster" throughout), removed redundant or inappropriate sentences, and refined terminology (e.g., replacing "expression" with "localization" for protein detection).

Figure Corrections: Fixed inconsistent Y-axis labels and numerical ranges (e.g., aligning percentages/probabilities with appropriate scales), resolved color scheme confusion, standardized oviposition-related labels to "Per female egg numbers within 3 days," and added details on sample sizes and replicates to all figure legends.

Statistical Improvements: Re-evaluated statistical analyses for proportion-based datasets (applying chi-square tests for contingency tables) and time-series data (using generalized linear models to address multiple testing), with revised methods clearly described in the text and figure legends.

Abstract Restructuring and Discussion Modification

Abstract: We have restructured the abstract to group results thematically (rather than sequentially) for improved readability. The revised abstract emphasizes the core findings: CRZ/*CrzR* signaling is critical for female PMR in both N. lugens and D. melanogaster, acts endogenously in females, and is required for male seminal fluid factors to induce PMR. Male-related content has been removed since experimental data are deleted from the rest of the paper.

Discussion: We have modified the discussion to include the evolutionary conservation of CRZ-mediated female PMR, the molecular and neurobiological implications of CRZ/*CrzR* signaling, and future research directions (e.g., dissecting downstream pathways in the female reproductive tract and brain). We have also reduced tangential content and clarified how our findings advance understanding of female endogenous signaling in PMR regulation. A new section was added at the end, which discusses outstanding questions related to CRZ and the PMR in both insect species.

To both the above-mentioned sections and the Introduction we also added new text to emphasize that CRZ is a paralog of the vertebrate peptide gonadotropin-releasing hormone (GnRH), a hormone known to regulate reproduction in vertebrates (including humans), thus suggesting conservation of an ancient role in reproduction.

All revisions in the manuscript are highlighted in red for easy reference. We believe these changes significantly strengthen the study's focus, clarity, and scientific impact. Thank you again for your time and consideration.

> ***Reviewer #1 (Recommendations for the authors):***
>
> *(1) The abstract could benefit from some restructuring. Right now, it reads like a sequential reporting of the results, but clumping together results thematically would make it easier to read, in my opinion. Also, see above re: my concerns about no evidence for the signal being endogenous in D. melanogaster.*

Thank you for your constructive suggestions regarding the abstract and the evidence for endogenous CRZ signaling in D. melanogaster. We fully agree with your feedback and have addressed both points thoroughly in the revised manuscript:

(1) Abstract Restructuring

We have restructured the abstract to group results thematically, rather than sequentially, to enhance readability and highlight the core findings. The revised abstract now organizes key information into three cohesive sections:

The context and significance of female post-mating response (PMR) regulation, emphasizing the gap in understanding endogenous female signaling pathways.

The core findings across both study species (Nilaparvata lugens and D. melanogaster), including the critical role of CRZ/CrzR signaling in suppressing re-mating and promoting oviposition, and its requirement for male seminal fluid factors to induce a PMR.

The conclusion regarding the evolutionary conservation of endogenous CRZ signaling in female PMR, reinforcing the study's broader implications.

We also added new text to emphasize that CRZ is a paralog of the vertebrate peptide gonadotropin-releasing hormone (GnRH), a hormone known to regulate reproduction in vertebrates (including humans), thus suggesting conservation of an ancient role in reproduction.

This thematic structure eliminates the linear "result-by-result" narrative, making the abstract more concise and impactful while clearly communicating the study's key contributions.

(2) Evidence for Endogenous CRZ Signaling in female D. melanogaster

To address your concern about the lack of evidence for endogenous signaling in female D. melanogaster, we have supplemented the manuscript with two sets of critical experiments confirming that CRZ is not derived from male accessory glands (MAGs) but acts endogenously in females:

RT-PCR Analysis: We performed RT-PCR on dissected male MAGs, male heads (positive control), and female tissues. Results showed no detectable *Crz* mRNA in MAGs, confirming that males do not synthesize CRZ in this tissue.

Immunohistochemical and Genetic Labeling: Using the GAL4–UAS system (*Crz*-Gal4/UAS-mCD8-GFP) to label *Crz*-expressing neurons, combined with anti-CRZ antibody labeling, we observed no *crz*/CRZ signal in male MAGs. This confirms that MAGs neither produce nor sequester mature CRZ peptide.

These findings demonstrate that CRZ signaling in D. melanogaster females is endogenous, as the peptide cannot be transferred from males during copulation. The new data are presented in Figure 5H-I and described in the Results section, with corresponding methods detailed in the Methods section.

The revised abstract integrates this new evidence to explicitly state the endogenous nature of CRZ signaling in both BPH and D. melanogaster females, aligning with the thematic structure and addressing your concerns comprehensively. We believe these changes significantly improve the clarity and rigor of the abstract and the manuscript overall.

> *(2) The authors use Drosophila as a broad placeholder throughout the manuscript, while they are specifically referring to D. melanogaster in several places. I would go through the manuscript and switch with the appropriate Drosophila species/species'.*

Thank you for pointing out this important detail regarding species-specific terminology. We fully agree with your suggestion to ensure accuracy and consistency in referencing the Drosophila species studied.

We have systematically reviewed the entire manuscript, including the abstract, introduction, results, discussion, methods, and figure legends, and revised all instances where the general term "*Drosophila*" was used. All references now explicitly specify "*D. melanogaster*" to accurately reflect the species utilized in our experiments.

> *(3) For the figures, I think the number of replicates is a distracting addition to the plot. This is still useful information, but could instead be added in as a line/table, in my opinion.*

Thank you very much for your suggestion. We have added the information on the number of replicates and sample sizes to the corresponding figure legends, which we hope improves clarity and readability.

> *(4) There are typos in the y-axis label of all of the oviposition figures. A better re-wording would be "Per female egg numbers within 3 days".*

Thank you very much for your suggestion. Following your recommendation, we have now standardized the Y-axis label for all oviposition-related figures to "Number of eggs per female within 3 days."

> *(5) In Figure 1B and Figure 1 - Supplement 3a, since the comparisons are solely between control vs treatment, I would not join means across treatments that I am not comparing.*

To address this, we have revised Figure 1B and Figure 1—Supplement 3a by removing the connecting lines between group means. The updated figures now display independent mean ± SEM values for each dose (Figure 1B) and time point (Figure 1—Supplement 3a), with significance markers only applied to the control vs. treatment comparisons we actually tested. This revision eliminates any implied relationships between non-comparative groups and ensures the data visualization aligns with our statistical approach. We appreciate the reviewer's suggestion, which has improved the clarity of the data presentation.

> *(6) The authors mention courtship rate in lines 511, but from a look at the methods, this is not the courtship rate! This is a measure of the number of males engaging in any form of courtship. Also, in Figure 5 Supplement 2A, it appears that under 1% of males are courting. This seems extremely low. Do the authors mean percentages? In that case, I would reformat from 0 to 100/relabel the y-axis.*

Thank you for your observation and valuable feedback on this terminology and figure presentation issue. We fully acknowledge the inaccuracies and have addressed them comprehensively:

(1) Correction of "Courtship Rate" Terminology

We agree that the term "courtship rate" in Line 511 was incorrect, as our measurement reflects the proportion of males engaging in any form of courtship (not a rate per unit time). However, since we have removed all male-related data (including this section and associated figures) from the revised manuscript to focus on the core finding of female post-mating response (PMR), this terminology error has been eliminated entirely.

(2) Revision of Figure 5 Supplement 2A

Consistent with the removal of all male-related experiments, Figure 5 and its supplementary materials (including Supplement 2A) have been excluded from the revised manuscript. This ensures the current work remains cohesive and centered on female PMR, while also resolving the Y-axis labeling ambiguity you identified.

We appreciate your careful attention to these details, which helps enhance the accuracy and clarity.

> *(7) It appears Figure 5A, 5D, and 5G are mislabeled? Aren't all rematings with wild-type males?*

Thank you for identifying this labeling inconsistency. You are absolutely correct, all re-mating assays in the original figures involved wild-type males, and the mislabeling was an oversight.

However, we have removed Figure 5 (and its associated subpanels A, D, G) entirely from the revised manuscript, as part of our decision to exclude all male-related data.

(8) I am not sure I understand why a 30-minute post-injection threshold was chosen and what this table means. Could the authors elaborate on the methodology here on how they quantified premature ejaculation?

> *Thank you for your question regarding the 30-minute post-injection observation window and the methodology for quantifying premature ejaculation.*

While we have removed all male-related data (including the corresponding table and premature ejaculation analyses) from the revised manuscript to focus on our core finding, this is no longer included in the manuscript.

> *(9) Line 29 - "distensible" seems an odd choice of word here.*

We have revised Line 29 and removed “distensible”. “Peptide injection and knockdown of CRZ expression by RNAi or CRISPR/Cas9-mediated mutagenesis demonstrate that CRZ signaling suppresses mating receptivity”.

> *(10) Line 57 - delete "a" from "a post-mating response" and "A PMR" because the authors are referring to a very specific suite of post-mating behaviours.*

We have revised Line 57 (and other relevant instances throughout the manuscript) to delete the article "a" from these phrases.

> *(11) Line 352, delete a from "and in a significantly".*

We have revised Line 356 to remove the extraneous "a", correcting the phrase to "and in significantly".

> ***Reviewer #2 (Recommendations for the authors):***
>
> *The work presented in this manuscript presents the study of the neuropeptide corazonin in modulating the post-mating response of the brown planthopper, with further validation in Drosophila melanogaster. To obtain their results, the authors used several different techniques, including dsRNA injection to induce RNA interference and CRISPR-CAS9-mediated site-specific mutagenesis. The experimental design is appropriate; the results are solid and support the conclusion of the manuscript. Overall, the merit of the manuscript is to present compelling evidence that the female post-mating response is mediated by corazonin, at least in the analysed species. There are multiple reports in multiple insect species, indeed, that male factors, particularly those secreted by male accessory glands, induce post-mating response in females, but the female pathways underlying this phenomenon are poorly understood.*
>
> *There are points the authors can consider to improve the manuscript quality.*

Thank you for your generous and insightful assessment of our manuscript. We deeply appreciate your recognition of the study’s strengths, including the appropriate experimental design, solid results, and meaningful contribution to understanding female endogenous pathways in post-mating response (PMR) regulation.

We have carefully incorporated all your constructive suggestions (e.g., statistical analysis revisions, figure label standardization, text refinements) to further strengthen the manuscript's rigor and clarity. By focusing on corazonin (CRZ/corazonin receptor (CrzR) signaling in female brown planthoppers (Nilaparvata lugens) and validating these findings in Drosophila melanogaster, we aim to provide a conserved model for female endogenous PMR regulation across insect species.

Thank you again for your thoughtful and supportive feedback, which has been instrumental in refining our work. We believe the revised manuscript now more effectively communicates the significance of CRZ-mediated female signaling in bridging the gap between male-derived cues and PMR execution.

> *(1) Line 20: "optimal offspring". This is not a zoological parameter. One can use "optimal fitness".*

We have revised Line 20 to replace "optimal offspring" with "optimal fitness" as recommended.

> *(2) Line 36-40: I think that the main message of the manuscript is the involvement of the corazonin pathway in controlling the female post-mating response. The involvement of corazonin in the male reproduction is also of note, but out of topic (in my opinion). The male corazonin is not transferred during mating from males to females, and the involvement of corazonin in controlling the gene expression in the MAGs is of note, but it is poorly related to the effect of corazonin in the female. I am not suggesting removing these data from the paper; they are important. But I do not find them that important to include them in the abstract, also because it confounds the reader at first. A similar statement can be made for the discussion (lines 728-745): making this the first piece of data commented on takes the stage, but this is not the main take-home message of the paper.*

Thank you for this suggestion. We fully agree that including male-related CRZ data in the abstract and leading the discussion with these results distracted from the primary focus and risked confounding readers. In fact, we also removed the entire section on the role of CRZ in males. We have addressed this issue comprehensively in the revised manuscript as follows:

(1) Abstract Revision

We have completely removed all content related to male CRZ function from the revised abstract. The updated abstract now exclusively emphasizes the core findings:

The requirement of CRZ/CrzR signaling for mediating key female PMR traits (suppression of remating, promotion of oviposition) in both *Nilaparvata lugens* and *Drosophila melanogaster*;

Experimental evidence confirming that CRZ acts as an endogenous female signaling factor (not a male-transferred molecule);

The evolutionary conservation of CRZ-mediated female PMR regulation across the two insect species.

We also added a comment on the evolutionary conservation of CRZ and GnRH signaling in reproduction.

(2) Discussion Section Restructuring

We have restructured the Discussion to prioritize the core message of female PMR regulation:

Lead paragraph adjustment: Lines 728–745 (originally focusing on male CRZ and MAG gene expression) have been deleted.

Revised opening focus: The Discussion now only contain a synthesis of our key findings on female CRZ signaling, including its molecular mechanisms, cross-species conservation, and implications for understanding endogenous female pathways downstream of male seminal fluid cues.

We appreciate your suggestions for the narrative focus of the manuscript.

> *(3) Line 49: "Reproductive behavior is critical for population sustenance and survival of the species": I find this intro a little teleological evolutionary speaking, and I am not totally sure that this has ever been demonstrated as a concept. I would skip it, simply saying "Reproductive behavior in insects is influenced...".*

Following your suggestion, we have revised Line 49 to streamline the introduction and avoid "teleological language". The updated sentence now reads: "Reproductive behavior in insects is influenced by a complex interplay of neural, hormonal, and environmental factors."

> *(4) Line 58: "A PMR has been documented across diverse insect taxa, including Drosophila melanogaster, Anopheles gambiae, Aedes aegypti, and the brown planthopper (BPH), Nilaparvata lugens". There are many other insect species for which PMR has been shown: crickets, fruit flies, grasshoppers, etc. Therefore, I would say "for example" to underline that it is not a complete list. Being an incomplete list, I suggest that the authors pay attention to the cited literature: the literature cited in the case of Anopheles gambiae demonstrates the synthesis of hormones in the MAGs, but it has nothing to do with PMR; there is nothing cited for Aedes aegypti, even if the authors named the species.*

Thank you for this constructive feedback on the framing of PMR studies across insect taxa and the accuracy of our cited literature. We fully agree with your suggestions and have addressed these issues comprehensively in the revised manuscript:

(1) Revision of the Sentence Structure

We have modified Line 58 to explicitly indicate that the listed species are examples rather than a complete inventory of insects with documented PMR. The revised sentence reads:

"The PMR has been documented across diverse insect taxa, for example, *Drosophila melanogaster*, *Anopheles gambiae*, *Aedes aegypti*, crickets (*Gryllodes sigillatus*), grasshoppers (*Dichromorpha viridis*), and *the brown planthopper (BPH)*, *Nilaparvata lugens*"

(2) Correction of Literature Citations

We have thoroughly reviewed the citations associated with the listed species to ensure they directly support the role of PMR:

For *Anopheles gambiae*: We have replaced the previously cited study (focused on MAG hormone synthesis) with two relevant references that explicitly characterize PMR traits—including mating-induced oviposition stimulation and remating suppression—in this mosquito species.

For *Aedes aegypti*: We have added two newly published studies that document key PMR phenotypes (e.g., post-mating refractoriness and altered feeding behavior) and their underlying molecular mechanisms in this species.

For crickets (*Gryllodes sigillatus*): We added a newly published study that documents PMR phenotypes in *Gryllodes sigillatus*.

We have also verified that the citations for *D. melanogaster* and *N. lugens* remain directly relevant to PMR regulation, with no adjustments needed.

All revised citations are properly formatted and integrated into the text, with corresponding updates to the reference list.

> *(5) Line 111-132: I find this redundant: it is a long summary of the methods and the results. I do not think it is needed here, but I think the authors should point to the main message of their data.*

Thank you for pointing out the redundancy of Lines 111–132. We fully agree that this section, disrupted the flow of the introduction of our study.

To address this, we have completely removed Lines 111–132 from the revised manuscript. In place of this redundant content, we have added a concise, focused paragraph that emphasizes the central hypothesis and key objective of our work: specifically, to identify the endogenous female signaling pathways that mediate the post-mating response (PMR) downstream of male-derived cues, and to validate the conserved role of corazonin (CRZ) signaling in this process across *Nilaparvata lugens* and *Drosophila melanogaster*.

> *(6) Line 156: This sentence is not needed here.*

We have deleted the sentence in Line 156 from the revised manuscript.

> *(7) Figure 1E, J supplementary 3A: The label of the Y axis is the percentage of the mating females (expected 0-100%), but the numbers show the fraction (0-1). On the contrary, in Figure 1 Supplement 4, the label says "probability of survival" and the probability goes from 0 to 1, while the number of the axis goes from 0 to 100 (percentage).*

Thank you very much for pointing out these inconsistencies. We have carefully reviewed all Y-axis labels and corresponding numerical ranges throughout the manuscript and corrected the mismatched axes.

> *(8) Figure1B, C, F, K supp 2, 3A: I found this use of colours confounding. Why did the authors use the light blue for sCRZ, but the mean and SE are shown in pink, which is the colour for CRZ? Furthermore, it is not reported anywhere how many individuals have been used per replicate. There is the total number of insects, the number of replicates, but there is no indication about the minimum number of insects per replicate in this and many other subsequent experiments.*

Thank you for identifying these critical inconsistencies in figure color coding and missing details on sample allocation per replicate, and we greatly appreciate your meticulous review of our data presentation.

We have addressed these issues in the revised manuscript as follows:

(1) Standardization of Color Coding

We apologize for the confusing color mismatch between group labels and data points in Figure 1B, C, F, K, and Supplements 2 and 3A. We have unified the color scheme across some figures to ensure consistency:

The *sCRZ* (control) group is now consistently represented by light blue for both labels and mean ± SE data points.

The *CRZ* (treatment) group is now consistently represented by pink for both labels and mean ± SE data points.

For Figures 1C, F, K and Supplementary Figure 2, we were concerned that the mean and s.e.m. bars might be visually obscured by the data points. To improve their visibility, we therefore used the opposite color to display the mean and s.e.m.

All figure legends have been cross-checked and updated to reflect this standardized color coding.

(2) Addition of Sample Size per Replicate

We acknowledge that the lack of information on the minimum number of insects per replicate was a key gap in our experimental reporting. We have supplemented this critical detail in this way:

Figure Legends: For Figure 1B, C, F, K, and Supplements 2 and 3A (as well as all subsequent experiments), we have added explicit statements specifying the minimum number of insects per replicate, alongside the total sample size and number of replicates (e.g., “n = 3 replicates, with a minimum of 10 females per replicate; total N = 35 females”). All revised figures and their corresponding legends have been integrated into the updated manuscript, and we have cross-checked all other figures to avoid similar issues.

> *(9) Figure 1C, F, K, Supplementary Figure 3B: Y axis labels - "Eggs numbers of per female...". I suggest changing it to "Number of eggs per female...".*

We have revised the Y-axis labels for Figure 1C, F, K and Supplementary Figure 3B to Number of eggs per female...” as recommended. Additionally, we cross-checked all other oviposition-related figures in the manuscript to ensure uniform use of this standardized label, eliminating any inconsistent phrasing across the dataset.

> *(10) Legend Figure 1B: Mann Whitney test. How did the authors perform the test? Hour by hour? I am not sure this is the best way to analyse the data, because it is a case of multiple testing. Probably a linear model or a glm might be a better fit.*

Thank you very much for pointing out this issue. In Figure 1B, each concentration group was analyzed using data from independent individuals, and therefore the comparisons do not involve repeated measures across time; for this reason, we consider the Mann–Whitney test appropriate for this dataset. For Figure 1—Supplement 3A, however, our original analysis compared treatment and control groups hour by hour, which indeed raises concerns regarding multiple testing. Following your suggestion, we have removed the potentially misleading connecting lines and reanalyzed the dataset using a generalized linear model (GLM). The updated figure and revised legend have been included in the revised manuscript.

> *(11) Legend Figure 1E: ANOVA test. These are proportions, not continuous variables of the samples. Tests for proportions might be a better fit (chi-square, etc.).*

To address this issue, we have re-analyzed the proportional data in Figure 1E using Pearson’s chi-square test of independence, which directly evaluates the association between treatment group (sCRZ vs. CRZ) and the binary mating status (mated vs. unmated) of females. This test is statistically robust for proportional data and avoids the assumptions of normality and homogeneity of variances required for ANOVA.

> *(12) Knockout experiments: I agree with the authors that the data are strong enough to sustain the conclusions. However, is the corazonin knockout haplosufficient or is it recessive? What is the behaviour of the heterozygotes?*

Thank you for this insightful question regarding the genetic basis of the corazonin (CRZ) knockout phenotype.

To address your query, we have supplemented experiments with additional phenotypic analyses of heterozygous CRZ knockout females (+/*ΔCrz*), and we clarify the genetic nature of the knockout as follows:

(1) Genetic basis of the CRZ knockout:

The CRZ knockout line was generated via CRISPR-Cas9-mediated deletion of the *Crz* coding region, resulting in a recessive loss-of-function mutation. Homozygous knockout females (*ΔCrz*) exhibited the full phenotypic suite reported in the manuscript (impaired post-mating suppression of remating, reduced oviposition rate, and disrupted CRZ signaling in the reproductive tract).

(2) Phenotype of heterozygous females:

Behavioral and physiological assays of +/*ΔCrz* heterozygotes revealed no significant differences compared to wild-type (+/*ΔCrz*) females across all measured post-mating traits. Specifically:

Remating rates of +/*ΔCrz* females were indistinguishable from wild-type controls at 48 h post-mating.

Oviposition output of +/*ΔCrz* females matched wild-type levels over a 3-day assay period.

(3) Updates to the manuscript:

We have added these heterozygote data as figure1J and K in the revised manuscript, with corresponding descriptions in the Results and Methods sections. We have also explicitly noted the recessive nature of the *Crz* mutation in the Genetic Manipulation subsection, ensuring clarity for readers.

These results confirm that the *Crz* knockout phenotype is fully recessive and that one functional copy of the *Crz* gene is sufficient to maintain normal post-mating responses—supporting our conclusion that CRZ signaling is required for mediating female PMR.

We thank you again for raising this important point, which has strengthened the genetic rigor of our study.

> *(13) Figure 1, Supplementary 1: I do not understand why the authors point out the fact that these are Protostomia. These are all Arthropoda, there is not a single species outside this Phylum. Caerostris darvini should be Caerostris darwini.*

Thank you for this feedback regarding Figure 1 and Supplementary Figure 1. We fully agree and have addressed these issues in the revised manuscript:

(1) Removal of the "Protostomia" designation

We have deleted all references to Protostomia from the figure legends and associated text.

(2) Spelling correction of *Caerostris darwini*

We apologize for the typographical error in the species epithet. We have corrected the misspelling *Caerostris darvini* to the taxonomically accurate Caerostris darwini (Darwin's bark spider) across all instances in Figure 1, Supplementary Figure 1, and their corresponding legends. We have also cross-checked all other species names in the manuscript to eliminate similar typographical errors.

> *(14) Line 299: CRZ expression: I found this confounding, given that the authors were talking about the expression of the gene. I would use the term localization, referring to the protein/peptide (is it what the authors were pointing at?).*

To resolve this ambiguity, we have revised Line 299 to replace *CRZ* expression with CRZ peptide localization, which accurately describes the experimental focus (immunofluorescence staining and confocal imaging of the CRZ protein). We have also cross-checked the entire manuscript to standardize this terminology:

We use *Crz* gene expression exclusively when referring to transcriptional analyses (e.g., qRT-PCR results).

We use *CRZ* peptide localization when describing the spatial distribution of the protein (e.g., immunostaining assays).

> *(15) Figure 2C: The expression is relative to...? I would make it explicit on the axis.*

Thank you for this helpful comment. We apologize that the normalization reference was not sufficiently clear in the original version. In the revised manuscript, we now explicitly state that RT–qPCR data were first normalized to the reference genes *Actin* and *18SrRNA*, and then expressed relative to the mean expression level of the tissue showing the highest *Crz* expression, which was set to 1. We have clarified this information in the figure legend and the Methods section.

We have revised Figure 2C as follows:

Updated the Y-axis label to explicitly state the reference: “Relative *Crz* gene expression”.

Added a supplementary note in the figure legend to confirm that relative expression values were calculated using the $2^{-\Delta\Delta Ct}$ method, with the reference gene serving as the internal control for normalization.

Additionally, we have cross-checked all other qRT-PCR-related figures in the manuscript to ensure that the reference for relative expression is clearly indicated on the corresponding axes, standardizing this key detail across all gene expression datasets.

> *(16) Figures 3B, E, I, L, M, N: Percentage and proportions, as in Figure 1; furthermore, please provide the minimum number of individuals per replicate. Furthermore, as in Figure 1, the data are proportions, and I would use statistical tests that are studied for this kind of data.*

Thank you for this helpful suggestion. We have reviewed and corrected the Y-axis labels and corresponding numerical ranges in these figures, and we have added the number of replicates and the minimum number of individuals per replicate to the figure legends. In addition, following your recommendation, we have reanalyzed these proportion data using chi-square tests for contingency tables.

> *(17) Figure 3: As in Figure 1, it would be interesting to know which is the behaviour of the heterozygotes.*

Thank you for suggesting to complement the data in Figure 3 with heterozygote phenotypic analyses.

To address this, we have conducted additional behavioral and physiological assays of heterozygous *CrzR* knockout females ($+/CrzR^{M}$) and integrated these data into the revised Figure 3 and its legend:

Phenotypic characterization of heterozygotes: Across all traits measured in Figure 3 (e.g., remating rate and oviposition efficiency,), $+/CrzR^{M}$ females exhibited no significant differences compared to homozygotes.

This confirms that the *CrzR* knockout phenotype is dominant and that one functional copy of the *CrzR* gene can't to maintain normal post-mating response (PMR).

Manuscript updates:

We added heterozygote data in Figure 3I and J. Accordingly, we updated the Results text to reflect the revised panel labeling.

We supplemented the figure legend with statistical comparisons between heterozygotes and wild-type groups (using chi-square tests for proportional data).

We included a brief description of heterozygote phenotypes in the Results section to contextualize the genetic basis of the *CrzR*-mediated PMR regulation.

> *(18) Figure 3 Supplement 1: Can the authors indicate which model for maximum likelihood they chose? Did they perform a pre-test to assess which substitution model was the best for their data?*

Thank you for this critical question regarding the model selection for maximum likelihood (ML) phylogenetic analysis in Figure 3 Supplement 1. We fully agree that specifying the substitution model and validation process is essential for ensuring the reproducibility and rigor of phylogenetic inferences.

To address this, we have supplemented the manuscript with detailed information on the model selection and validation steps, as follows:

(1) Substitution model selection

Prior to constructing the ML tree, we performed a model selection pre-test using the ModelFinder tool integrated in IQ-TREE 2, which evaluates the fit of candidate nucleotide substitution models to the *CrzR* amino sequence alignment via the Bayesian Information Criterion (BIC). The model selection procedure identified the LG+G model as the best-fit substitution model for our dataset. This model uses the Le and Gascuel (LG) amino-acid substitution matrix and incorporates a gamma-distributed rate variation among sites (G) to account for among-site rate heterogeneity.

(2) Manuscript updates

We have added this detailed model selection process and the final LG + G model specification to the legend of Figure 3 Supplement 1.

We have also included information on bootstrap validation (10000 ultrafast bootstrap replicates) to support the node support values reported in the phylogenetic tree.

> *(19) Figure 4 Supplement 1: I would be explicit about what it is relative to (which gene).*

Thank you for this helpful comment, In the revised manuscript, we now explicitly state that RT–qPCR data were first normalized to the reference gene *Actin*, and then expressed relative to the mean expression level of the tissue showing the highest *CrzR* expression, which was set to 1. This normalization strategy provides a robust and biologically representative reference. We have clarified this information in the figure legend and the Methods section.

> *(20) Line 518 and Line 525 and Figure 5: The authors show that injection of CRZ and RNAi of crz or mutant crz has the same effect on male fitness. How do the authors explain this*

> *contradiction? The CRZ injection should activate the pathway, and crz RNAi and mutant crz should inhibit the pathway, but nevertheless, they have the same effect. I would probably test the expression of some of the genes whose expression is altered in crz mutant males (next paragraph) to see if an altered CRZ signalling pathway (both ways) might affect gene expression in the MAGs in the same way.*

Thank you for raising this important point. As explained above, we have removed all data related to CRZ function in male BPHs from the current version.

> *(21) Figure 5, Figure 7: As in Figures 1 and 3, please pay attention to the percentages and proportions and the statistical tests.*

Thank you for pointing out these issues. We have carefully reviewed and corrected the percentage/proportion labeling in the relevant figures, including the Y-axis descriptions and numerical ranges, as well as revised the corresponding figure legends. In addition, we have reanalyzed the data using statistical tests appropriate for proportion data. All corresponding revisions have been incorporated into the updated manuscript.

> *(22) Line 728-745: As already stated for the abstract, the male effect of crz is, to me, a side product, and I am not sure the male crz signalling has something to do with the female crz signalling. It is interesting, nobody showed that CRZ affects expression in the MAGs, but this is not the main message of the paper, and it confuses the reader. I would reduce the discussion about this aspect and move it to the end, but this is my own take.*

We have removed all data related to CRZ function in males for the reasons outlined above.

> *(23) Material and methods/results: as a general suggestion, I would be explicit about the timing of receptivity inhibition in the species. I've seen the authors have established this in precedent work, and I would refer to that work and make the reader aware of how the receptivity works in the species (i.e., that it is not permanent and lasts for a few days after first mating). This allows a better understanding of the experimental design.*

Thank you for this valuable and constructive suggestion. We fully agree that explicitly describing the timing of receptivity inhibition in *Nilaparvata lugens*, and linking it to our earlier work, will strengthen the rigor and clarity of the manuscript.

To address this, we have revised the Materials and Methods and Results sections as follows:

(1) Materials and Methods (Experimental Design subsection)

We have added a dedicated paragraph that explicitly defines the temporal dynamics of post-mating receptivity inhibition in *N. lugens*, with direct reference to our prior work[1]. The text clarifies:

"In N. lugens, mating induces a transient suppression of female receptivity that is not permanent. Females typically start regain remating willingness 72 h after the first mating, as documented in our previous study[1]. This temporal window guided the design of our remating assays, in which females were paired with naive males at 48 h post-initial mating to capture both the suppressed and recovered phases of receptivity."

(2) Results (Post-mating Receptivity section)

We have incorporated a brief contextual sentence at the start of the section to reinforce this key species-specific trait, ensuring that readers connect our assay timings to the temporal dynamics of receptivity in *N. lugens*.

These revisions ensure that the rationale behind our experimental timing is transparent and well-supported, allowing readers to fully grasp how our assays were tailored to the biological characteristics of *N. lugens*.

> *(24) Line 854: There is a typo "CRZ peptide. virgin female", the dot should be a comma.*

We have revised Line 854 to correct the punctuation: the dot has been replaced with a comma, resulting in the phrasing "CRZ peptide, virgin female". In addition, we have changed the wording in this sentence to ensure scientific rigor and to avoid colloquial expressions.

(1) Zhang, Y.J., Zhang, N., Bu, R.T., Nässel, D.R., Gao, C.F., and Wu, S.F. (2025). A novel male accessory gland peptide reduces female post-mating receptivity in the brown planthopper. Plos Genet 21, e1011699. 10.1371/journal.pgen.1011699.

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